

# FRONT

Frontier Resurgent Operations  
Navigator Tracing

## The Unclosed Dependency

Correction at the Next-Cut Seam

The refusal can be closed.  
The dependency can remain open.  
Correction is a different operation.

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## Abstract

FRONT asks when retained, warranted work can replace rediscovery and when changed tasks still require new input, derivation or verification. Historical warrant, present applicability and future serviceability are different obligations. The Next-Cut Seam names the boundary of presently justified continuation, not an oracle for choosing or completing every next step.

FRONT also addresses an unsupported operative premise: its content can be contradicted while the event of its assertion, and its downstream uses, remain consequential. A finite correction contract preserves history, binds the challenge to the selected support, checks alternative derivations, and separates proposed remedies from permission, execution and revalidation. A source-derived Single-Cut example motivates the distinction without turning its static count into a temporal transport process. An elementary conditional invariant formalizes these obligations; FRONT does not claim invention of truth maintenance, belief revision or provenance tracking.

The FRONT-REUSE-01 baseline contains 27 before/after pairs. A distinct retained FRONT-CORRECT-01 study evaluates three policies on 75 synthetic fixtures and 216 support questions per policy. Selective checking and blanket rederivation answer all 216 correctly; unchecked continuation returns 114 invalid proof receipts, including 30 answers with alternative current support. A separate 96-case scratch transaction assay checks permission and secondary-constraint revalidation. Costs, slower cases, failure controls and limits are retained. The accompanying Lean audit checks 21 headed propositions: 17 at the stated general mathematical scope and four through explicitly bounded formalizations. No language-model, deployed-security or independent-referee result is claimed. [S17]

FRONT integrates scoped reuse, priced applicability, carried dependency, permission-aware correction and protected continuation. The formal companion records its assumptions and remaining bridges separately from the finite experiments. Independent review and public Single-Cut version matching remain open.

**Keywords:** warranted reuse; applicability cost; correction; dependency; provenance; continuation; permission; Single-Cut Transport; BRIDGE.

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# 1. Origin: the answer that should not need earning twice

The originating exchange did not start with a new SAT solver. It started with a familiar conversational problem: a system receives a meaningful “no,” carries on, and later behaves as though the same boundary must be discovered again. Pang’s observation was that some negatives really are settled within their scope. Understanding one should change future work, rather than merely produce an acknowledgment. [S0]

This follows a distinction already explicit in MIND v0.4: a negative outcome can be informative, a successful outcome can be informative, and repetition of an already known refusal under unchanged conditions need not add any information at all. The polarity of an answer is not a measure of its contribution. [S2, §4.3]

The conversation then reached a second question. Even a perfectly retained answer does not necessarily tell a solver which unresolved question matters next. Pang’s proposed shift was to treat that boundary not simply as a defect awaiting removal, but as a legitimate part of the account’s structure. **A completed local answer and an unfinished next task can coexist.** [S0]

That is FRONT’s center. It is neither a demand to keep every detail permanently in active memory nor a promise that a sufficiently large memory eliminates search. It asks what a retained result can actually do now, what its use costs, and where further work is genuinely needed.

## 1.1 Name and authorship

**FRONT — Frontier Resurgent Operations Navigator Tracing** was coined by **Christopher D. Pang**, including the acronym and its expansion, in the originating conversation. His first written expansion included the aside “Frontier(cheap).” The aside marks an interest in cost; it is not an extra word in the adopted expansion recorded in that exchange. Pang stated that he created the wording himself in roughly two minutes of forward-only composition, with no chatbot supplying the acronym or its component expansion. This account records that statement, not an independently timed cognitive measurement. [S0]

The assistant had previously proposed the label **Next-Cut Seam** and discussed related computational ideas. Those contributions are contextual ancestry, not authorship of FRONT’s name. The definitions, proofs, examples, literature synthesis, and implementation proposals in this manuscript were developed with AI assistance under Pang’s authorship and review. The entire manuscript is therefore not described as “100% human-written”; that provenance claim concerns the naming event.

## 1.2 What FRONT contributes

**The practical hinge is this: how much work can a system spend checking that old work still applies before reuse stops saving anything?** An inexpensive but unsound check admits stale results. A sufficient but overly conservative check discards useful ones. A fully semantic check can cost as much as solving again.

FRONT develops six connected parts:

1. A task-relative retained-warrant contract and Next-Cut-Seam account, distinguishing a supported answer, an adequate common action and a supported future continuation (§§3–9).
2. A full-cost account in which applicability, packaging, retrieval, reopening and recomputation have to be priced, rather than treating a cache hit as free (§11).
3. An indexed continuation adapter from BRIDGE: only declared and actually available views can support a recovery claim, and recovering the requested answer differs from recovering the entire state (§9; Appendix D).

4. FRONT-REUSE-01’s executed deterministic lane and frozen real-model protocol: a small repeat-and-change comparison with stale transfer, conservative misses and first-use costs reported separately (§16.1).
5. A history-preserving correction contract and FRONT-CORRECT-01, distinguishing failed selected support, alternative support, permission and revalidation (§§11.9 and 16.5; Appendix E).
6. A source-linked Lean companion with explicit assumptions, bounded formalization limits and reproducible proof receipts (§17.2).

**Claim boundary.** FRONT does not claim invention of caching, dependency tracking, change propagation, clause learning, protected-query reasoning or provenance-aware agent memory. Nor does FRONT prove a general complexity separation, a new Hodge result, a physical model of conversation, or measured LLM/token savings. The elementary and inherited mathematics keeps its source labels. The combined account is the contribution examined here; novelty of that integration remains open to further comparison. The current workload does not compare FRONT with a competitive incremental solver or mature build system.

### 1.3 The accidental zipper, and the question beneath the arrow

Pang connected repeated foaming and collapse in aerosol residue on a mirror with a fast white change he had watched in cooking bacon, and called the perceived pattern a zipper. These were reported observations and a resemblance, not measurements or evidence of a common physical mechanism. The moving-front interpretation and continuous equations were assistant proposals, not fitted explanations. [S8]

His two linked statements were “one structural ‘no’ can lead to a clean, atomic-promotion-free, chain of yesses” and “there’s no shortcut to figuring out a brand new ‘no’.” The first motivates propagation; the second asks what work precedes it. Section 6 retains the distinction between missing admitted input and a consequence present implicitly but not yet computed. [S8]

Pang’s next question, “How do you know what a no makes?”, shifts attention to **what holds up the therefore**: the other premises, the rule and its present applicability. The finite zipper in §11 is an assistant-selected realization of this question, not a physical theory of the original observations. Its purpose is to expose the rule basis without requiring every use to re-prove all of its ancestry.

### 1.4 A room of views, and a continuation of the same question

The later conversation returned to BRIDGE v0.4 through Pang’s phrase, “The residual jiggles.” BRIDGE already contained generator and readout changes and accumulated return. The new question was not how to animate a formerly static theory, but which of its presently hidden differences could remain consequential under continuation. Pang also recalled imagining a room from all its surface viewpoints while watching Feynman’s *Fun to Imagine* in his twenties, roughly fourteen years earlier. His correction was explicit: the idea was not merely more pictures, but “all of it at once”—the family and its relations as one object of consideration. [S9–S11]

Feynman’s radio example supplies a useful picture: changing the receiving instrument can make something readable that the previous instrument did not reveal. His discussion of why magnets attract supplies another: a familiar analogy can borrow the very relation it purports to explain. These are source-labeled motivations for checking an interface and its basis, not premises of FRONT’s mathematical results. [S10, supplied 42:50–43:13 and 21:04–21:36]

FRONT integrates that direction into its account of answer sufficiency (§7), decision sufficiency (§8), continuation (§9), and full cost (§11). It uses an indexed family of views and only the frame/continuation pairs specified by the account. A source-linked proof may make an earlier structural implication explicit; that work does not retroactively invent the architecture or observation that supplied the question. BRIDGE remains a separate, unchanged source. Reuse and applicability remain the organizing questions. [S11; S12]

### 1.5 The refused premise, and the work of correction

A later conversation revisited the supplied Single-Cut Transport note. Hearing “false spends upstream,” Pang distinguished an attempted configuration from an admitted one: a proposed use can be refused, and the refusal can

become informative for another continuation. He then pointed out that a refusal need not be respected. A rejected assertion can stay operationally influential without becoming valid. His subsequent formulation placed its unclosed consequences at FRONT’s frontier. A separate Hearthline episode was reported by Pang as identifying the next obligation: “Correction needs its own mechanism.” The present manuscript develops that proposal; it does not claim that the separate episode’s proof checks were independently recovered here. [S14]

The resulting addition is not a new definition of mathematical truth. It is a bounded correction contract: preserve the assertion event, identify the selected support that failed, check surviving alternatives, and obtain the appropriate permission before applying and revalidating a change. Pang supplies the originating observations and the v1.0 destination. The finite implementation, elementary proposition, experimental choices and editorial integration were developed with AI assistance. The unadopted v0.5 thinking key was planning ancestry, not an intervening FRONT release. [S14–S16]

## 2. Corpus placement and inherited limits

PAL v2.4 governs PAL terminology in FRONT. It describes an ordered mechanical account, not a universal mathematical theory. A receipt records an account; it does not make that account true or authorized. The first cut is supplied within an account, and closure is scoped and reopenable. [S1, §§1–4, 7]

MIND v0.4 supplies the question/fork, negation, attention, wake, and revision vocabulary. It already separates missing input from a consequence that has not yet been computed. CHARTER v1.0 supplies the fiber criterion, protected-query carriage, and the distinction between recomputing an answer and preserving its evidence history. These are same-author antecedents, not independent confirmation of FRONT. [S2, §§4–5, 8.7; S3, §§10, 15]

The Quiet Layer supplies a practical constraint: useful information need not become a new task simply because it is available. FRONT should help a run use its history, not create another ritual the run must satisfy before doing its work. [S4, “Core distinction,” “The ARC illustration”]

The Riemann–Pang Ledger and *The Price of Spherical Closure* provide motivating examples of local results and property-specific closure obligations. They do not supply a complexity-theoretic lower bound. Historical terminology in the spherical paper is retained as historical, not imported over PAL v2.4. [S5; S6]

BRIDGE v0.4 supplies the explicit detector/binding/promotion separation, typed endpoint sectors, query-repair law, and return comparison imported here. FRONT retains PAL v2.4 as its governing terminology; BRIDGE’s reference to PAL v2.3 remains historical. Its Hodge construction is not a premise or completion dependency of this revision. [S9]

Three separations are especially important:

**Unresolved residue is not  $\Omega$ .** PAL’s  $\Omega$  is metalinguistic and is not used here as a set, software value, frontier, oracle, random source, or reservoir of hidden answers. FRONT’s explicit unresolved register is a finite record of needs the account has identified. It does not enumerate all that is unknown. [S1, §§2–3]

**An answer is not a grant.** Mathematical impossibility, an empirical negative, a human refusal, and lack of evidence require different records. Knowing that an action is physically possible does not authorize it. Preserved knowledge does not renew expired permission. [S1, §§2, 6–7; S2, §§4.3–4.5]

**A status is not an answer.** PAL’s claim statuses remain `CLOSED_IN_SCOPE`, `OPEN`, `UNRESOLVED`, and `REOPENED`. A proved NO can be `CLOSED_IN_SCOPE`. A timeout is not such a NO. FRONT’s operational labels do not silently replace PAL’s four-status grammar. [S1, §7]

### 2.1 Assertion, falsity, and attributed intent

“Assertion,” “unsupported premise,” “contradicted premise,” and “lie” do not have the same evidentiary requirements. An assertion’s occurrence can be supported while its content is false; lack of support alone is not proof of falsity. A contradiction can be proved within declared premises, without establishing the speaker’s awareness or intent. Pang’s

explicitly nonliteral “simplest possible particle” analogy motivates a compact claim-event whose effects persist; it is not a physical theory or a theorem that deception is intrinsically unprovable. Social meaning and the authority to remedy an effect are additional questions. No human-exclusive theorem about meaning, or automatic machine authority, follows. [S14]

### 3. The room before the symbols

Imagine a workbench with four places. The first holds the current question. The second holds results already obtained, with their reasons attached. The third holds the resources and permissions currently available. The fourth is the unfinished work. Nothing requires the fourth place to be empty before a useful action can occur.

FRONT concerns movement between these places:

$$\begin{array}{ccc} \text{current task} & \rightarrow & \text{relevant retained result} \\ \text{result} & \rightarrow & \text{applicability check} \rightarrow \text{reuse or new work.} \end{array} \quad (1)$$

A successful reuse brings a supported result back into active service. An unsuccessful lookup merely fails to find a reusable result through that lookup. Neither event decides every later question.

#### 3.1 A bounded working account

Let the state at step  $t$  be

$$X_t = (I_t, T_t, K_t, U_t, B_t, G_t). \quad (2)$$

Here  $I_t$  is the admitted input and its relevant versions;  $T_t$  is the retained event-and-evidence trace;  $K_t$  is an active index of reusable results;  $U_t$  lists identified unresolved tasks;  $B_t$  is the current resource account; and  $G_t$  is the current external grant. These objects are supplied or maintained within a declared implementation. Equation (2) is not a claim that every reasoning system has six internal compartments.

The active index and the historical trace serve different purposes. An index entry can be invalidated or evicted without changing what was recorded at an earlier time. Conversely, preserving a long archive does not guarantee that a needed result can be found promptly. TIES’s separation between untouched ancestry and revisable, source-linked connections is a useful model for this distinction. [S7]

#### 3.2 A reusable result

Use a record

$$r = (q, \Gamma, v, \pi, \nu, D, \Sigma, \rho). \quad (3)$$

The question  $q$  fixes what was answered. The assumptions  $\Gamma$  fix the premises. The value  $v$  is the answer, when one was established. The object  $\pi$  is a proof, witness, observation record, or other evidence of the declared type. The identifier  $\nu$  names the checker and interpretation rules. The dependency set  $D$  names the relevant source versions. The scope  $\Sigma$  specifies where the result applies. The handle  $\rho$  locates the original trace and reopening conditions.

These fields can be references rather than copies. They are not all required as repeated natural-language instructions. A record for a simple pure function may need very little; a transported empirical conclusion may require substantially more.

Define the **resurgence** of  $r$  as its return to use after it was obtained earlier. Resurgence is not proof discovery, fresh measurement, or retroactive improvement of the original evidence. It is reuse under a current applicability basis.

### 3.3 The rule is retained structure too

A scoped inference can be written

$$\Gamma, d \vdash_R y. \quad (3a)$$

Here  $\Gamma$  contains the surrounding premises,  $d$  is a supplied or derived distinction,  $R$  identifies the applicable rules and interpretation, and  $y$  is the claimed consequence. The arrow is not powered by the polarity of  $d$  alone. Its basis includes the other premises and the rule system.

Equation (3) already has places for this basis:  $\Gamma$  for premises,  $v$  for the checker and interpretation,  $D$  for relevant versions,  $\Sigma$  for scope, and  $\rho$  for the trace. A rule or an applicability result can itself be referenced by an ordinary reusable-result record. No seventh state compartment or mandatory meta-controller is needed.

A trace can make the ancestry inspectable without making it self-validating. A bounded derivation ends at declared starting premises and rules; declaring those starting points does not establish that they are unquestionable foundations of reality. Their status remains available for a later, relevant challenge.

### 3.4 A view is part of the result's basis

An observation label is not decoration. For a retained decoder, the account must identify the state domain, the readout or frame, any preceding continuation, the requested query, and the relevant versions. These fit the existing fields: the question and premises specify the job; the interpretation identifier and dependencies specify the maps; the evidence and trace handle identify the readings actually obtained. A family of views may be stored by references rather than copied into every prompt.

BRIDGE distinguishes detection, binding, and account promotion. In its linear setting a residue has the form

$$\rho_f(x) = r_f((\Phi - \Phi_0)x), \quad (3b)$$

with the transformation, comparison, domain, and readout declared. A detector can establish that this output is nonzero. A binding is a sound implication to a *named* proposition; recording that checked proposition is a later account transition. A model's confidence, a familiar label, or a long history of repetitions does not supply the missing implication. [S9, "Detection Binding and Account Promotion"]

This also locates the conversational phrase *pre-promotion*: a process or readable effect can occur before a stronger interpretation has been established. That is a claim-relative distinction in the account, not an extra physical compartment. Conversely, an exact zero can be an earned answer. The receipt records what was established without making the underlying event wait for its paperwork.

### 3.5 Carriage is not admission

The six-place account already represents the new case. The event "p was asserted and used" belongs in T with its recorded basis; it need not enter I as established content. If an unsafe imported cache treated p as warranted, correction withdraws that selected support from K, while preserving its historical use in T and opening the affected requalification work in U. Resources remain in B and external grants in G. There is no seventh mandatory compartment or fifth PAL status.

A diagnostic label is not a substitute for current support. A receipt's hash identifies bytes; it does not authenticate the underlying source's truth or authority. Alternative derivations must carry their own source and applicability witnesses. The prototype's trusted/available/version fields are supplied synthetic inputs, not an algorithm for deciding which humans or observations to trust. Conditional reasoning from p can remain a conditional artifact after affirmative reuse of p is disqualified. [S16]

## 4. A hard NO is an answer, not a general-purpose negative flag

For a Boolean decision problem, distinguish YES and NO from the operational outcome UNRESOLVED. Also keep ERROR, NOT\_APPLICABLE, and NOT\_AUTHORIZED separate when the implementation needs them. These are declared outcome labels; their spelling alone provides no evidence.

An exhaustive negative proof answers a question. An interrupted search reports how far the procedure got. A rejected permission request reports a boundary on action. Combining these into a single false-like value invites the system to misread one kind of event as another.

The practical counterpart is simple. “The package did not import” is not “the puzzle is unsolvable.” “This checker ran out of time” is not “the proposition is false.” “The user declined” is not an invitation to seek a different route to the same effect.

A fresh query should reuse an answer only after establishing the relevant applicability conditions. That does **not** require re-asking a person for an unchanged grant on every turn. A mechanical grant can remain in force until its declared expiry, revocation, or material scope change. Nor does it require rediscovering a theorem whose proof and premises are already available. Rechecking applicability and re-solving the original problem are different operations.

### Proposition 1 — Scoped deductive reuse

**Elementary consequence of ordinary deduction.** Fix a sound monotone deductive calculus with admissible substitution of premise derivations, such as ordinary propositional natural deduction. Suppose a retained proof establishes  $\Gamma \vdash \varphi$ , and the present account  $\Delta$  establishes every premise in  $\Gamma$ . Then

$$\Delta \vdash \varphi. \quad (4)$$

**Proof.** For each use of a premise from  $\Gamma$ , insert its derivation from  $\Delta$ , retaining the proof’s rules and substitutions. The resulting derivation proves  $\varphi$  from  $\Delta$ . This is a standard proof-composition argument. Its validity does not give a bound on the size of the composed proof or the time needed to find the premise derivations. ■

A particularly cheap sufficient condition is exact premise inclusion,  $\Gamma \subseteq \Delta$ , under unchanged interpretations. A more general entailment check may be expensive. FRONT must not hide that expense behind the word *applicable*.

Cook and Reckhow explicitly allow natural-deduction proofs to share already derived formulas under the same assumptions rather than repeat the derivation. The reuse principle is established proof practice, not a FRONT invention. [R2, §3]

For defeasible observations, changed physical conditions, or modified software, (4) is not a license to preserve the answer blindly. Their dependence on circumstances must be represented in the premises and versions, or evaluated by a separate, adequate transfer rule. Dependency-maintenance systems are direct prior art for this problem. [R5; R6]

### 4.1 A NO works with its background restriction

Consider the finite-domain inference

$$x \in \{0, 1\}, x \neq 0 \Rightarrow x = 1. \quad (4a)$$

The same exclusion on  $\{0, 1, 2\}$  leaves both 1 and 2. The NO did not carry the whole answer: the domain premise did indispensable work. A cached conclusion that forgets that premise will misapply an otherwise valid inference.

This example also distinguishes kinds of basis. An observed input is a record of what was observed under stated conditions. A declared domain or definition fixes the exercise. A hypothesis supports a conditional conclusion. A grant authorizes an action within its scope. None silently turns into another because a subsequent calculation is coherent. Permission to construct a model does not establish that the model describes the mirror, the bacon, or any other external system.

“Atomic-promotion-free” is retained here as Pang’s conceptual phrase, not a new formal definition of PAL. Its operational use in this paper is limited: accumulating rule applications does not by itself upgrade a hypothesis into an observation, an analogy into an established mechanism, or a grant into a truth certificate. The model in §11 keeps those distinctions visible without importing a new ontology.

#### 4.2 A rejected proposal is neither a spent allowance nor a repair

An attempted SCT orientation is data evaluated by an admissibility predicate. Testing it costs computation even when the proposed arrangement is refused. That computational expenditure is not the row’s internal-exclusion charge. Likewise a permission NO and a technical failure must remain different: another command spelling does not create permission for the same denied effect. Neither kind of NO supplies a successful repair or guarantees a return to a valid state. [S15–S16]

## 5. A small SAT laboratory: when a NO becomes reusable structure

Let  $a, b, c$  be Boolean variables, and consider

$$F = (\neg a \vee b) \wedge (\neg a \vee c) \wedge (\neg b \vee \neg c). \quad (5)$$

Trying  $a = \text{true}$  forces both  $b$  and  $c$  to be true. The third clause rejects that combination. Instead of retaining only “this attempt failed,” derive a clause that explains the failure:

$$\begin{array}{ll} (\neg a \vee b), (\neg b \vee \neg c) & \vdash (\neg a \vee \neg c), \\ (\neg a \vee c), (\neg a \vee \neg c) & \vdash \neg a. \end{array} \quad (6)$$

Each line is one resolution step. The result rules out all four assignments with  $a$  true, not merely the particular ordering in which the solver encountered the contradiction.

However,

$$\text{Models}(F) = \text{Models}(F \wedge \neg a). \quad (7)$$

The learned clause does not remove real solutions. It makes an implication of the original formula explicit and can spare subsequent search. The example has three models:  $a$  is false, and  $b, c$  are not both true. Its eight assignments and the two proof steps are checked again by the new v0.2 companion described in §15.

Conflict analysis, learned clauses, and nonchronological backtracking are precisely the territory of GRASP and subsequent clause-learning SAT research. FRONT uses this established mechanism as a concrete model of an earned negative returning as useful structure. [R4]

Incremental SAT under assumptions is an even closer operational comparison: related calls reuse solver state and learned clauses, while assumptions mark what changes between calls. *Ultimately Incremental SAT* explains how assumption dependence must be represented before learned information can safely cross calls. FRONT’s example is a diagnostic laboratory within that established territory, not a new incremental SAT algorithm. [R16, §1]

## Proposition 2 — Negative-core transfer

**Elementary SAT fact.** If a set of clauses  $C$  is unsatisfiable and  $C \subseteq F'$ , then  $F'$  is unsatisfiable.

**Proof.** A satisfying assignment of  $F'$  would satisfy every clause of  $C$ , contradicting its unsatisfiability. ■

The corresponding rule for a positive witness is different. An assignment satisfying  $F$  need not satisfy  $F \wedge D$ . Added constraints preserve an UNSAT core but may invalidate a SAT witness. Deleting clauses can do the reverse. “Same kind of problem” is not a transfer certificate.

### 5.1 A stale answer that manufactures a false NO

Let

$$F_0 = a \wedge (\neg a \vee b). \quad (8)$$

From  $F_0$ , derive  $b$ . Now remove the premise  $a$  and add  $\neg b$ :

$$F_1 = (\neg a \vee b) \wedge \neg b. \quad (9)$$

The assignment  $a=b=false$  satisfies  $F_1$ . But a cache that carries  $b$  into the new account without its dependency check obtains  $b \wedge \neg b$  and incorrectly announces UNSAT.

The historical proof of  $b$  from  $F_0$  remains correct. Its application to  $F_1$  is unsupported. Keeping these two statements separate is more informative than either deleting the old proof or treating it as permanently valid everywhere.

The reconstructed companion enumerates all 256 formulas made from the eight nonempty, non-tautological clauses over two variables. All 64 formulas containing the original two premises admit the cached derivation, and its conclusion holds in every model of those formulas. The explicit stale case demonstrates why this check matters. These counts describe that finite family only.

### 5.2 A learned NO with a genuine downstream chain

The original three-variable example earns  $\neg a$ ; it does not determine  $b$  and  $c$  individually. To study a subsequent chain, add three fresh variables and explicitly supply its links:

$$F^* = F \wedge (a \vee z_1) \wedge (\neg z_1 \vee z_2) \wedge (\neg z_2 \vee z_3). \quad (9a)$$

The two-step resolution derivation (6) still supplies  $\neg a$ , because its three premises remain present. With that seed, successive synchronous unit-propagation passes give

$$\{\neg a\} \rightarrow \{\neg a, z_1\} \rightarrow \{\neg a, z_1, z_2\} \rightarrow \{\neg a, z_1, z_2, z_3\}. \quad (9b)$$

Each arrow applies a supplied clause to the previous snapshot. The first added clause forces  $z_1$  when  $a$  is false; the other two force  $z_2$  and  $z_3$ . The original three choices of  $(b, c)$  survive, so  $F^*$  has exactly three models. Adding the sound learned unit  $\neg a$  does not alter that model set. Without the seed, unit propagation finds no initial unit clause and returns a stable partial assignment; it has not decided the formula. Conflict-based learning and Boolean constraint propagation are established SAT mechanisms. [R4]

Remove the middle link  $(\neg z_1 \vee z_2)$ . The same seed still forces  $z_1$ , but the sequence stops there. The three satisfying choices for  $(b, c)$  combine with the three satisfying choices for  $(z_2, z_3)$  under  $\neg z_2 \vee z_3$ , giving nine models. A procedure that can no longer derive  $z_3$  must not report  $\neg z_3$ ; both values occur in surviving models.

The companion enumerates all 64 assignments to each six-variable formula and records the actual waves. Its deliberately simple scanning procedure examines six clauses on each of three productive passes and a final no-change pass: 24 clause checks. Removing the link leaves five clauses and requires two passes: 10 clause checks. These counts exclude the prior resolution work, clause construction, and proof bookkeeping. They describe this implementation, not an optimal SAT cost.

The original derivation remains a historical fact after a link is removed. Its reuse for a descendant is what must be reconsidered. The dependency tests in §11.5 check rule content and versions as well as premises, and they allow an alternative surviving proof to recover a conclusion whose earlier proof no longer applies.

### 5.3 When a refused premise stays in use

The local Single-Cut Transport example fixes a finite order and a fully selected corridor. Its exact count is  $\kappa = e + s$ , with admissibility  $\kappa \leq 1$ . The supplied Boolean convention sets  $e = 1$  for false and  $e = 0$  for true. At a clause endpoint requiring  $s = 1$ , the first orientation has charge 2 and is inadmissible; the second has charge 1 and is locally admissible if the remaining constraints allow it. “False” here is an orientation, not a dishonest act. A puncture inside the corridor instead contributes another actual zero,  $\kappa = e + 1 + s$ , and is not the same phenomenon as refusing a proposal. The local theorem is unchanged. [S15, §§3–5]

Now let  $p$  abbreviate the assertion that the refused arrangement is valid. A participant or unsafe importer records  $p$  as accepted and derives  $q$  through a rule requiring  $p$ . The assertion and use really occurred. They do not establish  $p$ , and the selected derivation cannot currently warrant  $q$ . Repeating the assertion one hundred times increases the recorded history, not the row’s capacity, the proposition’s support, or the actor’s grant. This is an operationally accepted bad premise, not an additional interpretation of the static counting theorem. [S16]

Suppose  $q$  also has a recorded derivation requiring both  $a$  and  $b$ , and both roots remain supported. Rechecking that conjunctive derivation can retain  $q$  and its subsequent consequence  $z$ . It cannot validate  $p$  or retrofit the failed route. If no recorded alternative survives, the procedure returns no current recorded support; it does not infer  $\neg q$ . A different root version supporting the same conclusion is a distinct case from an independent alternative derivation and is counted separately.

Section 5.1 already shows invalid transfer of a once-supported cached result. This example adds no rival logic of inconsistency. Its additional workload is correction after a bad premise has remained in use: bind the challenge, retain historical dependency edges even when the current graph changed, requalify descendants individually, and separate any authorized state change from the evidence that proposed it. Section 11 specifies that operation; §16 tests it. The source’s all-false OR case also supplies a concrete warning: rejection need not leave a satisfying alternative at all. [S15–S16]

## 6. The Next-Cut Seam is not one undifferentiated ignorance

**Definition.** Relative to a task, retained interface, procedure, and resource budget, a **Next-Cut Seam** occurs where the retained account no longer provides an adequate continuation through the available justified reuse route. Crossing it requires a further supplied distinction, a further derivation, a changed representation, or an additional choice whose basis must be named.

The word *adequate* belongs to the task. It does not mean globally best, uniquely determined, emotionally satisfying, or guaranteed to settle all future questions. A program may have several equally valid next steps. A scientist may sensibly choose among several experiments. A completed task may require no further cut at all.

Four cases must remain distinct.

**Input seam.** The admitted evidence aliases alternatives that require different answers. No processing of that same input alone can reveal which alternative is actual.

**Access seam.** The answer is determined by admitted information but has not been computed, retrieved, or checked within the budget. A complete SAT formula belongs here, not automatically in the first case.

**Transfer seam.** A retained result has a known historical basis but no established applicability to the new question, versions, or environment.

**Choice seam.** Several continuations are available, and a task-specific criterion, experiment choice, or tie-breaking convention remains to be supplied. This can be cheap and need not obstruct progress.

These are diagnostic distinctions, not mandatory interface stages. A real event can involve more than one. Discovering that a cache is stale may expose an access problem; reading a source may resolve an input problem but leave a choice. The purpose is to avoid claiming one kind of resolution from another kind of result.

## 6.1 The full-input counterexample to an overbroad seam argument

Let the admitted input be the exact formula itself:  $\tau(F) = F$ . Then the SAT answer factors through  $\tau$  trivially. In fact it is computable by a finite exhaustive search. Whether a uniform algorithm always obtains it in polynomial time is the unresolved complexity question. [R1]

Thus “the answer was not known to this run” does not mean “the answer is absent from the mathematical input.” Likewise, an inability to predict an external observation proves nothing about the time required to decide an explicitly encoded formula.

MIND already protects this distinction: deterministic postprocessing cannot separate external alternatives that its complete input aliases, yet reasoning can make previously uncomputed consequences accessible. FRONT relies on both halves. [S2, §5.4]

## 6.2 Three questions at the leading edge

For a proposed zipper, ask separately: how was the seed obtained; what establishes the rule’s present applicability; and what must be computed once both are available? These questions locate work within the existing seams rather than introduce a mandatory fifth seam. A missing measurement can be an input seam, an uncomputed proof an access seam, and an old rule used under changed conditions a transfer seam.

“Brand new” can therefore refer to different events. A distinction that the complete observed interface cannot separate needs additional distinguishing input for an always-correct answer. A consequence not yet found by a particular procedure may already follow from its mathematical input. The latter observation supplies no general lower bound on the fastest possible algorithm.

Neither acquisition nor propagation is universally the expensive part. A seed can be cheap to obtain; a long propagation or expensive applicability test can dominate. The purpose of v0.2 is to keep these bills separate, not to assume the desired inequality.

## 6.3 Changed views use the existing seam distinctions

A missing observation can be an input seam; an uncomputed prediction from fully supplied data can be an access seam; a changed readout, label convention, or update can create a transfer seam; and choosing an adequate next probe can be a choice seam. These are applications of the existing distinctions, not a mandatory fifth stage. A coordinate change alone may leave the available information unchanged. Sections 7–9 separate that case from an additional observation.

## 6.4 Correction crosses existing seams

A correction can expose an input seam when needed evidence is absent, an access seam when a valid alternative has not yet been derived, a transfer seam when a recorded proof’s basis fails, or a choice seam when multiple admissible

repairs require a decision. These remain the existing four diagnostic distinctions, not compulsory workflow stages. A settled contradiction may coexist with an open consequence, an unchosen remedy, or an unauthorized action. Detection does not discharge those obligations. [S16]

## 7. Answer sufficiency: what the retained carrier can distinguish

Let  $H$  be a supplied set of admissible histories,  $\tau : H \rightarrow Z$  a retained representation, and  $q : H \rightarrow Y$  a total protected query. Histories with the same retained value form a **fiber** of  $\tau$ . Define

$$EqKer(\tau) = \{(h, h') : \tau(h) = \tau(h')\}. \quad (10)$$

For clarity beside the linear realization below, this revision writes  $EqKer$  for the pair relation that v0.2 denoted by  $ker$ . The mathematical statements and equation identifiers are unchanged.

### Proposition 3 — Fiber criterion for an answer

**Inherited criterion, restated.** A decoder  $d : \mathfrak{I}\tau \rightarrow Y$  satisfying  $q = d \circ \tau$  exists exactly when

$$EqKer(\tau) \subseteq EqKer(q). \quad (11)$$

**Proof.** If  $q = d \circ \tau$ , equal retained values give equal answers. Conversely, if  $q$  is constant on each fiber, define  $d(z) = q(h)$  for any  $h$  with  $\tau(h) = z$ . Fiber constancy makes this well-defined. ■

This is CHARTER’s Corollary 10.4 in FRONT notation, not a new theorem. It concerns a function on the reachable image. It does not by itself make that function continuous, computable, efficient, or authorized. [S3, §§10.1–10.4; S1, §2]

For a finite protected family  $Q$ , retaining every answer requires

$$EqKer(\tau) \subseteq \bigcap_{q \in Q} EqKer(q). \quad (12)$$

When queries are partial, answerability must also be preserved; two unavailable answers do not establish an answer. One can encode definedness explicitly, but an equality of “unknown” labels then certifies only the status, not the missing content. [S3, §15.3]

### 7.1 A result and its ancestry are different queries

Suppose two histories produce the same answer but use different proofs or source versions. An answer-only cache merges them. A protected query asking which source was used separates them. No decoder can recover that source identity from the answer alone if the identity varies within an answer fiber.

CHARTER states this distinction explicitly: cheaper recomputation of the mathematical claim need not preserve the evidence-bearing carrier. FRONT therefore separates a performance cache from the trace needed for later attribution, review, or reopening. [S3, §§10.4, 15.2–15.3]

This does not command universal retention. It makes the cost of a chosen loss visible. A compact summary can be sufficient for one declared task while inadequate for another.

## 7.2 The indexed family is one interface

The single-carrier criterion extends directly to several views. Let  $S$  be a common state set,  $F$  an index set, and  $o_f: S \rightarrow Y_f$  a declared readout for each frame. Define

$$J_F(s) = (o_f(s))_{f \in F}. \quad (12a)$$

### Proposition 3A — Indexed joint sufficiency

**Elementary application of Proposition 3.** The equality relation of the joint readout is

$$EqKer(J_F) = \bigcap_{f \in F} EqKer(o_f). \quad (12b)$$

A total protected query  $q: S \rightarrow Q$  has a decoder on  $\mathfrak{J} J_F$  exactly when this intersection is contained in  $EqKer(q)$ .

**Proof.** Two indexed tuples are equal exactly when their entries agree at every index. This proves (12b); Proposition 3 then gives the decoder criterion. The decoder is defined only on reachable tuples, where the value of  $q$  is independent of the representative. ■

The family is relationally indexed, not an unlabeled bag. For  $S = Q^2$ , the views  $o_x(x, y) = x$  and  $o_y(x, y) = y$  distinguish  $(1, 0)$  from  $(0, 1)$ . Forgetting the labels and keeping only the multiset  $\{0, 1\}$  merges them. Reordering fully labeled records does not make that loss. An arbitrary tuple also need not come from a common state:  $(1, 3)$  is incompatible with the joint readout  $(x, 2x)$ .

This is the bounded mathematical reading of “all of it at once.” It does not assert that simultaneous acquisition has more information than the same complete indexed observations gathered serially from a stable source. A changing source requires time labels and a justified comparison or update model. A perceptual image, a joint mathematical object, and a feasible measurement procedure remain different claims. [S11]

A relevant optical neighbor is the light-field representation of Levoy and Hanrahan: under their static-scene, fixed-illumination, unobstructed-space assumptions, images sample a four-parameter family of rays. FRONT uses that work as a representation comparison, not as an identification of a person’s imagined percept or proof of complete scene recovery. [R13, §§1–2]

## 7.3 A related view need not be recoverable from this view

For the linear realization, fix a finite-dimensional space  $T$  over  $Q$  and linear readouts  $r_f: T \rightarrow R_f$ . Their kernels are subspaces, unlike the equality relations denoted by  $EqKer$  above.

### Proposition 3B — View-to-view factorization

**Elementary linear factorization.** A linear map  $D_{g \leftarrow f}: \mathfrak{J} r_f \rightarrow R_g$  satisfying  $r_g = D_{g \leftarrow f} r_f$  exists exactly when

$$\ker r_f \subseteq \ker r_g. \quad (12c)$$

**Proof.** Factorization makes every vector killed by  $r_f$  also killed by  $r_g$ . Conversely, define  $D_{g \leftarrow f}(r_f x) = r_g x$ . The inclusion makes this independent of the choice of  $x$ ; linearity follows from the two linear maps. ■

Thus  $x$  determines  $2x$ , but on unrestricted  $Q^2$ ,  $x$  does not determine  $y$ . A common source and related view maps do not supply an image-to-image oracle. In a non-vector application, use the fiber criterion instead of assigning a linear kernel to a photograph or unknown geometry.

An invertible coordinate change is different again. If  $P : T \rightarrow T'$  and  $C : R \rightarrow R'$  are invertible and  $r' = C r P^{-1}$ , then  $r'(P x) = C r x$  and  $\ker r' = P(\ker r)$ . The same distinction is expressed in new coordinates; no new observation was acquired. BRIDGE's frame-transport account requires the relevant commuting maps rather than treating a change of notation as a new witness. [S9, "Frame Morphisms and Certificate Transport"]

## 8. Decision sufficiency: unresolved alternatives may share a next move

The originating formulation was too strong when it spoke as though a useful account must determine a unique next cut. A decision does not always require resolving all differences in the world. What matters is whether the unresolved differences change the set of adequate choices.

Fix finite nonempty sets of histories  $H$  and actions  $A$ , a retained map  $\tau : H \rightarrow Z$ , and a nonempty set  $G(h) \subseteq A$  of adequate actions for each history. Adequacy is stipulated by the current decision contract, including relevant outcomes, budgets, and grants. It is not inferred from a model's confidence. For example, "wait" cannot be added to every  $G(h)$  unless waiting actually satisfies the task.

### Proposition 4 — Common-continuation criterion

**Proposed FRONT formulation of an elementary set fact.** There is a deterministic rule  $p : \mathfrak{I} \tau \rightarrow A$  such that

$$p(\tau(h)) \in G(h) \text{ for every } h \in H \quad (13)$$

if and only if every retained-value fiber has a common adequate action:

$$\forall z \in \mathfrak{I} \tau, \bigcap_{h: \tau(h)=z} G(h) \neq \emptyset. \quad (14)$$

**Proof.** If  $p$  exists, its single action  $p(z)$  lies in every adequate set in that fiber. Conversely, choose one action from each nonempty finite intersection and define  $p(z)$  to be that action. The chosen rule satisfies (13). ■

This is an existence criterion, not a cheap algorithm for discovering the sets  $G(h)$  or their intersections. In a large problem, computing the acceptable-action sets can contain the original problem's difficulty. No novelty is claimed for the underlying finite-set reasoning.

### 8.1 Two different kinds of unfinished

Suppose a controller does not know whether a room is slightly too warm or slightly too cool. Under a generous tolerance, the adequate sets might be

$$G(h_1) = \{\text{wait}, \text{heat}\}, G(h_2) = \{\text{wait}, \text{cool}\}. \quad (15)$$

The temperature distinction remains unresolved, but waiting is adequate in both. No new measurement is required for that particular decision.

Change the task to immediate correction with no tolerance, and the adequate sets might become  $\{\text{heat}\}$  and  $\{\text{cool}\}$ . Now the intersection is empty. The same retained evidence no longer supports a universally adequate terminal decision. A measurement could be a justified *information-gathering* action, but its cost and its role in a longer policy must be evaluated separately.

This is the operational version of Pang’s seam: preserving an unresolved boundary need not block useful work, while hiding a boundary that changes the action can cause an error. **A seam becomes relevant through the job asked of the retained structure.**

## 8.2 A quantitative version

For finite real-valued loss  $L(h, a)$ , define the worst-case excess loss of choosing from a fiber by

$$\epsilon^*(z) = \min_{a \in A} \max_{h: \tau(h)=z} \left[ L(h, a) - \min_{b \in A} L(h, b) \right]. \quad (16)$$

A common action is within  $\epsilon$  of the history-specific optimum exactly when  $\epsilon^*(z) \leq \epsilon$ . This follows by substituting  $G_\epsilon(h) = \{a : L(h, a) - \min_b L(h, b) \leq \epsilon\}$  into (14). It does not supply the losses, their units, or a justified tolerance. Those are additional task inputs. The companion checks the exact set criterion, not a numerical optimization implementation of (16).

## 8.3 Visibility and task relevance are different axes

In a linear account let  $M : T \rightarrow Z$  be the available present interface. Let  $P$  be a specified family of protected query/continuation pairs  $(q, w)$ , with linear update  $L_w$ . A difference  $d$  is visible now when  $M d \neq 0$ , and relevant to these answers when  $q L_w d \neq 0$  for at least one requested pair.

| Present distinction | Some requested answer differs           | No requested answer differs                                    |
|---------------------|-----------------------------------------|----------------------------------------------------------------|
| Readable now        | Available and relevant to this contract | Visible but irrelevant to these answers                        |
| Hidden now          | Presently unresolved, yet relevant      | Ignorable for these answers after the criterion is established |

Every requested answer is determined by  $M$  exactly when

$$\ker M \subseteq \bigcap_{(q, w) \in P} \ker(q L_w). \quad (16a)$$

This is Proposition 3 in linear form, applied to the protected family. It does not make the differences intrinsically worthless or safe to delete from an archive: source attribution, a changed task, and later reopening may ask different questions. Nor does answer recovery replace Proposition 4: a common adequate action may be enough even when not every answer is known.

## 8.4 Stopping and correcting remain task-relative

No recorded proof of  $q$  is not itself a proof of  $\neg q$ , and inability to find a repair is not proof that none exists outside the candidate set. A task may permit STOP, conditional reporting, or requesting a grant; the common-action criterion must actually include that choice before it is called adequate. In the quiet fixtures, the task explicitly requires no further work, so STOP is correct. The example does not grant a solver universal permission to abandon obligations merely by renaming the stop Home. [S16]

## 9. Continuation sufficiency: tomorrow may ask a different question

An answer sufficient for the present question may not support the next input. To describe that distinction, fix a deterministic transition system with state set  $S$ , finite input alphabet  $A$ , total transition function  $\delta: S \times A \rightarrow S$ , and protected output  $o: S \rightarrow Y$ . Let  $\delta^*(s, w)$  apply a finite input word  $w$ , including the empty word.

Define future-output equivalence by

$$s \equiv_o s' \Leftrightarrow \forall w \in A^*, o(\delta^*(s, w)) = o(\delta^*(s', w)). \quad (17)$$

This is the familiar state-equivalence idea behind automata minimization: the retained distinction must survive relevant future inputs, not only the present display. [R10, Lemma 9.2 and equations 9.11–9.12]

### Proposition 5 — Future-output preservation

**Standard continuation argument.** For a compression  $c: S \rightarrow Z$ , every query  $q_w(s) = o(\delta^*(s, w))$  has a decoder from  $c(s)$  exactly when

$$EqKer(c) \subseteq \equiv_o. \quad (18)$$

**Proof.** Apply Proposition 3 to each  $q_w$  and intersect the resulting kernel conditions. ■

Decoder existence for each word is not yet an efficient online implementation. To update the *same chosen compressed representation* step by step, a sufficient and necessary factorization condition for an exact  $\hat{\delta}$  satisfying  $c(\delta(s, a)) = \hat{\delta}(c(s), a)$  is

$$c(s) = c(s') \Rightarrow c(\delta(s, a)) = c(\delta(s', a)) \text{ for every } a. \quad (19)$$

Condition (18) alone need not make an arbitrary representation satisfy (19). Taking the quotient by  $\equiv_o$  does provide a well-defined transition: appending one input preserves future-output equivalence. The quotient is a mathematical object; its construction and execution still require resource bounds.

For an explicit counterexample to confusing (18) with (19), take three states  $r, s, t$ , all with constant protected output zero. Let  $c(r) = c(s) = 0$ ,  $c(t) = 1$ , and for a single input  $a$  set  $\delta(r, a) = r$ ,  $\delta(s, a) = t$ ,  $\delta(t, a) = t$ . All future protected outputs agree, so (18) holds. Yet the two states encoded as zero require different next encodings, violating (19). A simpler one-state representation would suffice for the constant-output task. The extra encoded distinction is not automatically a useful one.

### 9.1 A six-bit demonstration

Let  $s \in \{0, 1\}^6$ . The current task asks only whether at least one bit is set. The one-bit summary

$$c(s) = s_1 \vee s_2 \vee \cdots \vee s_6 \quad (20)$$

answers perfectly. But a later task may ask for any coordinate  $s_i$ . The summary merges 63 distinct nonzero strings. Every pair of different strings can be separated by at least one coordinate query.

## Proposition 6 — Cost of protecting all coordinate queries

**Elementary counting bound.** An exact fixed-length encoding that answers every coordinate query for every  $d$ -bit string without external rereading requires at least  $d$  bits.

**Proof.** If two distinct strings had the same code, a coordinate on which they differ would violate Proposition 3. The encoding must therefore have at least  $2^d$  distinct codewords. A  $b$ -bit encoding supplies at most  $2^b$ , so  $b \geq d$ . ■

For the six-bit fixture, the current query uses two classes, while the full family of future coordinate queries requires 64. The reconstructed companion checks all 1,953 distinct pairs merged by the OR summary and finds a distinguishing query for each.

A pointer to an intact archive can avoid keeping six bits in the *active* summary. It does not refute the bound: the decoder then has external access. The resource account must include the archive and rereading route. This is why a short retained handle and a short self-sufficient representation are different claims.

## 9.2 Continuation-sensitive residue across admitted views

The preceding continuation criterion fixes one output. We now let the account vary both the view and the continuation. Retain a common domain  $S$ , the transition system  $\delta^*$ , and readouts  $o_f: S \rightarrow Y_f$ . Specify

$$A \subseteq F \times W, J_A(s) = (o_f(\delta^*(s, w)))_{(f, w) \in A}. \quad (20a)$$

The index set names the frame/continuation pairs in this interface. It is not automatically the full Cartesian product. **A pair's inclusion in a proposed measurement contract does not establish that its value has been obtained.** A run using only a collected subset must use that subset's actual interface in its decoder claim. Equally, an allowed action and an available observation are different records.

The formulas below use total maps on a common domain. A partial continuation can instead return a tagged definedness result together with its value. An unavailable reading is not the number zero, and omitting an entry without its status can erase a distinction. Mutually exclusive branches of an unknown physical state cannot be collected merely by writing them into a tuple. [S11]

### Proposition 5A — Combined-interface sufficiency

**Application of Propositions 3 and 3A.** A retained map  $\tau: S \rightarrow Z$  supports every output in (20a) exactly when

$$EqKer(\tau) \subseteq \bigcap_{(f, w) \in A} EqKer(o_f \circ \delta_w), \delta_w(s) = \delta^*(s, w). \quad (20b)$$

For finite-dimensional linear  $T$ , declared linear  $L_w: T \rightarrow T$ , and linear  $r_f: T \rightarrow R_f$ , put

$$J_A x = (r_f L_w x)_{(f, w) \in A}, K_A = ker J_A = \bigcap_{(f, w) \in A} ker(r_f L_w). \quad (20c)$$

A linear protected query  $q: T \rightarrow Q$  factors through  $J_A$  exactly when  $K_A \subseteq ker q$ . If  $A \subseteq A'$ , then  $K_{A'} \subseteq K_A$ .

**Proof.** Apply the equality-of-tuples argument to the composed readouts. In the linear case, a tuple is zero exactly when every component is zero, giving the kernel intersection. Define the query decoder by  $J_A x \mapsto q x$ ; it is well-defined precisely under the stated inclusion. Adding components intersects with further kernels and can only shrink the common kernel. ■

The factorization describes what the supplied readings determine, not how to acquire them or choose the cheapest ones. The new application of the existing sufficiency criterion is the explicit indexed contract: a difference can be

hidden in the present view yet distinguished by a particular later view. **Detection of one difference still does not guarantee recovery of every possible cause.**

### 9.3 A hidden change becomes readable

Appendix D.5 retains the exact delayed-readout example (20d)–(20e): under a known update, three accumulated scalar readings recover a three-component state, whereas one reading does not. A different update preserves invisibility indefinitely. These are demonstrations of what the chosen interface distinguishes, not a source of free future observations.

For FRONT’s main argument, the next example is the more direct test: **can one actually obtainable extra reading repair the answer we need?** It links combined-interface sufficiency (5A) to query-specific repair (5E), without requiring full recovery or the complete BRIDGE calculus to follow the practical question.

### 9.4 A changed view detects an effect without identifying its source

Use a different update and two possible views:

$$L(x, y, z) = (x, y+z, z), r_x(x, y, z) = x, r_y(x, y, z) = y. \quad (20f)$$

The difference  $d = (0, 0, 1)$  is missed by  $r_x$ ,  $r_x L$ , and  $r_y$ , but  $r_y L d = 1$ . Relative to the initial  $x$ -view, both a change of view and a continuation are needed to detect this chosen difference.

Suppose the actual observation family contains only  $(x, \emptyset)$  and  $(y, 1)$ . Then

$$J_A(x, y, z) = (x, y+z), K_A = \text{span}\{(0, 1, -1)\}. \quad (20g)$$

The states  $(0, 1, 0)$  and  $(0, 0, 1)$  produce the same actual readings  $(0, 1)$ . Their  $z$ -values differ. The observation detects a specified  $z$ -directed change, yet does not determine  $z$  for an unknown initial state. This is an explicit input collision, not a failure of calculation.

One additional reading— $y$  before the update—repairs this query, because  $z = y_{\text{after}} - y_{\text{before}}$ . The other missing pair,  $x$  after the update, is redundant. Including the entire rectangular family would recover the state too, but only by adding data absent from the original account; the query does not require both missing pairs.

This example keeps the next-cut question concrete. The work is not “seek more context” without a target. It is to identify a distinguishing observation, establish its availability and basis, and charge for obtaining it.

### 9.5 When invisible differences stay invisible

Fix  $T = Q^n$ ,  $n \geq 1$ , a known linear update  $L$ , and a fixed readout  $r$ . Define

$$O_N x = (r x, r L x, \dots, r L^{N-1} x), K_N = \bigcap_{j=0}^{N-1} \ker(r L^j), N \geq 1. \quad (20h)$$

#### Proposition 5B — Persistent invisibility and finite-horizon closure

**Standard observability argument, with proof in Appendix D.1.** Writing  $K = \ker r$ , the following are equivalent on the full linear difference domain:  $L(K) \subseteq K$ , and  $r L^j d = 0$  for every  $d \in K$  and every  $j \geq 0$ . Moreover,

$$K_{N+1} \subseteq K_N, K_\infty := \bigcap_{j \geq 0} \ker(r L^j) = K_n. \quad (20i)$$

The subspace  $K_\infty$  is  $L$ -invariant, and two states have equal outputs for all iterations exactly when their difference lies in it. The stacked map, Cayley–Hamilton stabilization, and invariance are established observability mathematics; FRONT’s application connects them to the retained-interface contract. [R12, lecture pp. 19–4–19–7, 19–12]

Restricting admissible states can weaken the necessary condition: only differences of actually admitted pairs matter. Nor does equality at sampled times establish equality for every continuation unless the model and argument justify that conclusion. For several supplied updates and all finite words, the invisible space is invariant under each update by concatenation. A restricted pair set need not have this closure property.

Keeping the history also differs from keeping the latest value. A coordinate swap on  $Q^2$ , starting at  $(0, 1)$  and reading the first coordinate, gives  $0, 1, 0$ . The last zero does not erase the retained middle distinction.

## 9.6 A finite witness for a fixed family—not free access to all views

### Proposition 5C — Finite subfamily with the same hidden space

**Elementary finite-dimensional argument.** Let  $\dim T = n$  and let  $r_f : T \rightarrow R_f$  be any indexed family of linear maps with finite-dimensional codomains. A subfamily of at most  $n$  readouts has the same common kernel as the whole family. The empty subfamily suffices when every readout is zero.

**Proof.** Appendix D.6 retains the finite row-span argument. ■

For example, every member of  $r_a(x, y, z) = x + a y$ ,  $a \in Q$ , misses  $z$ ;  $r_0$  and  $r_1$  already have the family’s complete row span. **All these views still do not recover all the state.**

The proposition is about existence in a fixed finite-dimensional family, including the combined readouts  $r_f L_w$  when its hypotheses apply. It does not find the useful probes in an infinite or black-box presentation, bound their acquisition costs, or extend automatically to noise and nonlinear or infinite-dimensional models. Section 11.7 keeps those costs in the same account as reuse.

## 9.7 What BRIDGE adds: visibility relative to a generated span

The kernel describes what the interface cannot distinguish. BRIDGE adds a second, independent question: what has the declared export generated? Fix  $e : G \rightarrow T$ , write  $B = \Im e$ , and use the combined readout  $J = J_A$  with  $K = \ker J$ . The four sectors are canonical subquotients, not a chosen decomposition into four complementary subspaces. [S9, “Four Sector Endpoint Theorem”]

| Sector | Subquotient      | What it records           |
|--------|------------------|---------------------------|
| $C$    | $B \cap K$       | Generated, but invisible  |
| $G_v$  | $B / (B \cap K)$ | Generated and visible     |
| $H$    | $K / (B \cap K)$ | Ungenerated and invisible |
| $V$    | $T / (B + K)$    | Ungenerated and visible   |

### Proposition 5D — BRIDGE at the combined interface

**Specified instance of BRIDGE’s existing endpoint theorem.** There is a natural exact sequence

$$0 \rightarrow \frac{K}{B \cap K} \xrightarrow{i} \frac{T}{B} \xrightarrow{j} \frac{J(T)}{J(B)} \rightarrow 0, V \cong \frac{J(T)}{J(B)}. \quad (20j)$$

Here  $i([k]) = [k]$  and  $j([x]) = [Jx]$ . In finite dimension,  $\dim T = \dim C + \dim G_v + \dim H + \dim V$ . Appendix D.2 supplies the maps and proof, including compatibility under readout refinement.

Holding  $B$  fixed while adding observations changes visibility, not the generative defect  $T/B$ . In example (20d), take  $B = \text{span}(1, 0, 0)$ . At horizons one, two, and three, the sector dimensions  $(C, G_v, H, V)$  are respectively  $(0, 1, 2, 0)$ ,  $(0, 1, 1, 1)$ , and  $(0, 1, 0, 2)$ . The total generative defect remains two-dimensional while more of it becomes visible.

There must also be **one common generated baseline across the views**. Let  $T = Q^2$ ,  $B = \text{span}(1, 1)$ , and  $J(x, y) = (x, y)$ . The joint quotient  $J(T)/J(B)$  retains  $x - y$ . But each separate quotient  $r_x(T)/r_x(B)$  and  $r_y(T)/r_y(B)$  is zero. Quotienting each coordinate independently has allowed different generated baselines and erased the cross-view relation. It is a different interface, not another notation for (20j).

## 9.8 Repair the requested answer, not necessarily the whole state

A query  $q: T \rightarrow Q$  asks about the state itself. A query  $q_D: T/B \rightarrow Q$  asks about the generative residual class; it corresponds to a state query annihilating  $B$ . These domains must not be substituted for one another. For the second query, the hidden obstruction is the embedded  $i(H)$ , not an untyped “residual.”

### Proposition 5E — Query-specific side-trace capacity and minimum rank

**Application of BRIDGE’s repair argument.** For finite-dimensional  $T$ , a linear query  $q$ , readout  $J$ , and side space  $E$ , a linear side trace  $s: T \rightarrow E$  makes  $q$  decodable from  $(J, s)$  exactly when

$$\ker J \cap \ker s \subseteq \ker q. \quad (20k)$$

Such an  $s$  exists exactly when  $\dim E \geq d_q$ , and its minimum possible rank is

$$d_q = \text{rank}(q|_{\ker J}). \quad (20l)$$

For a residual-target query replace  $T, J, \ker J$  by  $T/B, \acute{J}, i(H)$ , giving debt  $\text{rank}(q_D \circ i)$ . Appendix D.3 gives the lower bound and an attaining extension; the choice is not canonical. [S9, “Protected Answers and Minimal Repair”]

In (20d), the  $z$ -query debts at horizons one, two, and three are  $1, 1, 0$ . At horizon one, appending  $z$  settles that query while  $y$  remains hidden. A sufficient answer can therefore be strictly cheaper in information than full state recovery. Whether the extra measurement itself is available or economical is a separate question. Minimum rank is not minimum price, bit length, or operator effort.

## 9.9 A quotient needs compatible dynamics; return needs a comparison

### Proposition 5F — When an update descends to a quotient

**Elementary well-definedness criterion.** For a subspace  $B \subseteq T$ , the formula  $[x] \mapsto [Lx]$  defines a linear update on  $T/B$  exactly when  $L(B) \subseteq B$ . The analogous update on  $T/K$  requires  $L(K) \subseteq K$ . Appendix D.4 gives the proof.

The stacked readout exists without the first condition. In (20d),  $B = \text{span}(0, 1, 0)$  is not  $U$ -invariant because  $U(0, 1, 0) = (1, 1, 0)$ . Zero and  $(0, 1, 0)$  represent the same source quotient class but their images do not. A static BRIDGE quotient is lawful here; that proposed quotient update is not. This is the linear counterpart of FRONT’s earlier separation between output sufficiency and online-update compatibility.

BRIDGE already supplies accumulated return in its branch-indexed monodromy setting. Suppressing those indices locally, its identity is

$$R_k = (M^k - I)\alpha, R_{k+\ell} = R_k + M^k R_\ell. \quad (20m)$$

The second term is transported, not merely added as an unchanged remainder. The algebra follows from  $M^{k+\ell} - I = (M^k - I) + M^k(M^\ell - I)$ . A readout can hide a nonzero return discrepancy; a fixed class has exact zero return. Example (20d) illustrates the algebra but is not claimed to be monodromy of a geometric family. BRIDGE’s cycle-lifting and Hodge obligations remain at their original scope. [S9, “Monodromy Return”]

These results return to FRONT’s starting problem: carry enough supported structure to answer the next question, but do not turn an observed difference, a chosen representative, or a static quotient into an unearned continuation.

### 9.10 A corrected answer can retain a different history

Two continuations may agree on a repaired answer while differing on who relied on the old assertion, which source version was used, or whether an effect occurred before correction. Equality of the present answer therefore need not preserve a later ancestry query. The existing continuation criterion applies to these protected queries too: retained evidence must separate whichever alternatives the declared future task distinguishes. This is an application of the criterion, not a new BRIDGE theorem. A scratch rollback is particularly limited: it restores the modeled dictionary state, not the past assertion event or an irreversible external consequence. [S16]

## 10. No free oracle emerges from retained trace

Suppose two admissible external histories produce the same complete observation available to an agent, but their true answers to a requested query differ. A deterministic agent receiving only that observation must return the same output in both cases, so it cannot be correct in both. This is Proposition 3’s contrapositive, already present in CHARTER’s declared-input ceiling. [S3, §15.1]

Independent randomness does not guarantee a correct answer either. For two aliased histories with opposite binary answers, let  $p$  be the probability that the algorithm outputs YES. Its success probabilities are  $p$  and  $1 - p$ , so its worst-case success is at most  $1/2$ . A correlated side signal would change the available information and therefore the premise.

This is a genuine information limitation in the declared interface. It is **not** a proof of  $P \neq NP$ . If the complete SAT instance is in the input, the ambiguity argument does not apply to its truth value merely because a particular procedure has not computed it.

Likewise, a semantic summary written by a language model cannot recover source distinctions lost from all of its inputs. It may generate a plausible account, propose a hypothesis, or identify a useful next query. Those are not the same as reconstructing which original event occurred. FRONT records the difference between a proposal and a checked reopening.

### 10.1 Structure does not require endless refusal

An unresolved future does not prevent a local result from being final within its scope. It also does not establish that a proposed question has no answer. FRONT needs neither “everything can be solved now” nor “something must remain unknowable in every task.” Both would overspend the evidence.

The stronger working sentence is narrower: **carry what has been established, and do not manufacture the continuation that has not.**

## 10.2 A specified interface is not an acquired observation

The map  $J_A$  describes the observations in an account, not a device for manufacturing their values. In (20g), the known equation  $y_{after} = y + z$  and a measurement of that sum do not reveal either summand alone. Reading the update again does not supply the missing  $y$ -value. Likewise, permission to collect a view is not evidence that it was collected. Apply sufficiency to the actual supplied representation, including its labels and definedness; charge new acquisition when the interface changes.

# 11. The bill: when reuse actually saves work

A reuse claim needs a baseline and units. For one result used  $m \geq 1$  times, consider a deliberately simple model. Let  $D$  be the work to derive an answer from scratch;  $B$  the extra one-time packaging and indexing cost; and  $R$  the cost of each later lookup, matching, applicability check, verification, and retrieval. Assume the relevant conditions remain unchanged and all three costs use the same units.

$$C_{fresh}(m) = mD, C_{reuse}(m) = D + B + (m - 1)R. \quad (21)$$

## Proposition 7 — Amortized reuse condition

**Elementary cost identity.** Under these assumptions, reuse is cheaper exactly when

$$(m - 1)(D - R) > B. \quad (22)$$

**Proof.** Subtract the two expressions in (21) and rearrange. ■

If  $D \leq R$  and  $B \geq 0$ , the model gives no saving. For  $D = 10$ ,  $B = 30$ ,  $R = 2$ , four uses cost 40 without reuse and 46 with it; five uses cost 50 and 48. These stipulated-work-unit values are checked by the script. They are not timing measurements.

Real systems have variable costs, invalidation events, many queries, and storage pressure. A fuller account records acquisition, derivation, indexing, matching, checking, retrieval, update, and retention costs separately. Unknown costs stay unknown. A measured energy value is not interchangeable with elapsed time or operator effort without a declared comparison rule. CHARTER's Systemic Friction profile supplies this separation. [S3, §15.2]

The old literature is instructive here: de Kleer's ATMS discussion warns that straightforward bookkeeping can dominate the solver, and reports operations for which caching costs more than recomputation. Preserving structure has a price even when its logical utility is clear. [R6, §6]

## 11.0 Applicability is the priced hinge

Proposition 7 is an accounting identity, not an algorithm for making its repeat cost small. Its practical content depends on the price and error profile of the applicability test. Treat **false reuse**, **missed reuse**, and **necessary recomputation** as different outcomes. A failed sufficient test says that this route has not justified transfer; it does not establish that the old answer is false.

In a changing workload, charge the stages actually executed: lookup; reading and hashing relevant sources; validating a witness or dependency relation; reopening evidence; recomputing a rejected or insufficient answer; and packaging the new record. Time, source accesses, arithmetic operations and storage are separate dimensions. First-use costs belong in the total, even when later turns look cheap. Where two stages share one source read, charge it once and record the reuse of those already-read bytes.

Exact source equality is an economical sufficient condition only under a declared deterministic interpretation and adequate dependencies. It can reject equivalent reordered data. A retained proof can sometimes justify more: an UNSAT core that remains present survives strengthening without re-solving the whole formula. Which of these checks is cheaper than derivation is a workload question. Build traces and self-adjusting computation already address closely related overheads; the present study is a transparent instance, not a replacement for them. [R14–R16]

The applicability mechanism also has an upstream cost. A server-provided version stamp, index or dependency graph has to be created and maintained somewhere. The deterministic study therefore reads and hashes the current file rather than assuming a free freshness oracle. The real-model protocol additionally requires that initial model mistakes and certificate verification be priced. A fluent answer alone is not a reusable warrant. [S13]

## 11.1 Quiet reuse rather than a new compulsory loop

The question is not “did the model visibly review every stored fact?” It is whether the next action or answer appropriately uses a relevant retained result. An unchanged scope can be handled by an index and dependency check; it need not become a speech about permission. A failed retrieval need not start an unbounded sequence of self-checks.

A small integration should therefore attach to existing task flow:

```
A current task becomes relevant.
Look for an applicable retained result within the retrieval budget.
If one is available, use it and retain a reference to its basis.
Otherwise, let the existing solver act, compute, ask, or stop as appropriate.
Record material outcomes and changes; do not manufacture a verdict.
```

This is an optional design sketch, not a requirement that every token or action pass through a new semantic controller. The Quiet Layer remains the governing reminder about attention: support should not become another task that displaces the task being supported. [S4]

## 11.2 A finite propagation realization

Let  $L$  be a finite, explicitly represented vocabulary of facts, with  $N = |L|$ . Let  $R$  be a finite set of grounded rules  $A_r \rightarrow b_r$ , where  $A_r \subseteq L$  and  $b_r \in L$ . Rules, interpretations, and relevant versions remain fixed during one run. These facts are tokens with declared meanings; the spelling `not_a` alone is not an automatic negation calculus.

For  $S \subseteq L$ , define the one-step operator

$$\Phi_R(S) = S \cup \{b_r : A_r \subseteq S, r \in R\}. \quad (22a)$$

Let  $E$  be the initially explicit facts, and  $d \in L$  a seed whose basis has been declared. Set

$$E_0 = E \cup \{d\}, E_{j+1} = \Phi_R(E_j), W_{j+1} = E_{j+1} \setminus E_j. \quad (22b)$$

The sets  $W_j$  are the successive newly exposed facts. They may contain several facts, so this is a logical wave, not an assertion of a spatially contiguous physical front. Full closure is defined below as the eventual fixed point; repeatedly applying a completed closure would hide the steps we are trying to account for.

### Proposition 7A — Finite stabilization and least closure

**Elementary finite-rule argument.** For the supplied  $L, R, E_0$ , the sequence reaches a fixed point after at most  $N - |E_0|$  strict rounds. Its final set is the least  $R$ -closed superset of  $E_0$ , denoted  $cl_R(E_0)$ .

**Proof.** Every step retains the previous set. A strict step adds at least one member of the finite complement of  $E_0$ , so there can be at most  $N - |E_0|$  such steps. Equality at one step gives equality thereafter, since the operator is fixed. If

$T$  contains  $E_0$  and is closed under every rule, induction on  $j$  gives  $E_j \subseteq T$ : every premise of each new conclusion is already in  $T$ , so its conclusion is too. The final set is closed and contained in every such  $T$ , proving leastness. ■

The bound counts strict rounds, not all operations. The companion performs one final no-change pass as well. With  $m = |R|$  and  $\ell = \sum_r |A_r|$ , a scan that examines every rule and every body occurrence on each pass records exactly  $(k+1)m$  rule checks and  $(k+1)\ell$  premise membership tests for  $k$  productive rounds. The implementation separately constructs dependencies and copies sets; those operations are not included in the two counters. Finding a suitable rule representation, its size, initialization, matching, and storage remain part of the larger bill.

This is a finite forward-rule construction, not a claim to a new general closure method. In the concrete positive-rule interpretation used by the exhaustive fixture, one can check its least result independently by enumerating all Boolean valuations that satisfy the rules and initial facts and intersecting their true-fact sets. This separate finite oracle checks the implementation without being proposed as an efficient general solver.

### 11.3 Conditional soundness and declared starting points

#### Proposition 7B — Sound propagation relative to a basis

**Elementary induction.** Fix an interpretation and an admissible situation  $h$ . Suppose every fact in  $E_0$  is true in  $h$ , and every rule in  $R$  preserves truth in  $h$ : whenever all members of  $A_r$  are true,  $b_r$  is true. Then every fact exposed by (22b), and hence every fact in its final closure, is true in  $h$ .

**Proof.** The initial facts are true by assumption. If all facts in  $E_j$  are true, every rule fired at the next step has true premises and a true conclusion. Thus all facts in  $E_{j+1}$  are true. Apply induction and Proposition 7A. ■

If a member of  $E_0$  is hypothetical, the statement applies only to situations satisfying that hypothesis along with the other premises and rule assumptions. It does not make the hypothesis actual. Nor does the theorem establish that any such situation exists. The companion carries hypothetical-root dependencies into its descendants and rejects a grant object supplied as a factual premise; these are software distinctions, not verification of the world described by a premise label.

An empty-body rule is allowed. Its conclusion is an explicitly supplied axiom of the chosen rule account, not a discovery from no basis. A cyclic rule graph can also be allowed; it need not start itself. For  $p \rightarrow q$  and  $q \rightarrow p$ , no initial facts and no empty-body rules give an empty closure. Supplying  $p$  activates  $q$ . The distinction is between a declared root and an unsupported cycle, not between graphs with and without any cycles.

For the positive grounded fixtures, tokens have the ordinary Boolean valuation interpretation. General semantic interpretations can include incompatible facts; soundness remains conditional on the premises. The separate SAT fixture returns an explicit conflict on contradictory units and does not add ordinary prospective conclusions from the conflicting pass. No universal inconsistency-management system is inferred from these two small implementations.

### 11.4 What appeared after the seed, and what depended on it

The facts first exposed in a seeded run need not all owe their accessibility to that seed. Some may already follow from the original explicit facts and rules but simply have not been computed. Write

$$B = cl_R(E), S = cl_R(E \cup \{d\}), \Delta_d = S \setminus B. \quad (22c)$$

Here  $\Delta_d$  is the seed-dependent difference in final rule closure. It includes  $d$  itself when  $d \notin B$ ; remove  $d$  to count only additional consequences. This is a comparison of two fixed rule accounts, not an automatic causal claim about an external experiment. Computing both closures is work, not a free attribution operation.

**Proposition 7C — Decomposing the exposed wake**

**Elementary set consequence.** Let  $X_d = S \setminus (E \cup \{d\})$ , the union of the productive waves in (22b). Then

$$X_d = ((B \setminus E) \setminus \{d\}) \dot{\cup} (\Delta_d \setminus \{d\}). \quad (22d)$$

If  $d \in B$ , then  $S = B$  and  $\Delta_d = \emptyset$ .

**Proof.** Monotonicity gives  $E \subseteq B \subseteq S$ . Partition  $S$  into the disjoint sets  $B$  and  $S \setminus B$ , then remove  $E$  and  $d$ . This gives (22d); a fact first exposed at a step is never removed, so the productive waves have union  $X_d$ . If  $d \in B$ , the closed set  $B$  already contains  $E \cup \{d\}$ ; leastness gives  $S \subseteq B$ , and monotonicity gives the reverse inclusion. ■

For example, take rules  $p \rightarrow q$  and  $r \rightarrow s$ , initial facts  $E = \{p\}$ , and seed  $d = r$ . The seeded run exposes  $\{q, s\}$  in its first wave. But  $B = \{p, q\}$ , and the closure difference is  $\{r, s\}$ . The seed-dependent consequence is  $s$ ;  $q$  was unfinished prior work, not an effect unique to the new seed.

This also sharpens the SAT example. The sound learned  $\neg a$  is already a semantic consequence of  $F^*$ , so adding it does not change the models. It is nevertheless unavailable to the initially stalled unit-propagation procedure. A stronger derivation makes that consequence explicit and allows the restricted procedure to continue. **New explicit access, new final rule closure, and new semantic information are different changes.** Scheduling or lookup cost may change even when the final closure does not.

**11.5 A proof may be reusable without justifying itself**

For each first-discovered fact, retain one chosen derivation node. Initial facts have depth zero. A rule applied to a previous snapshot produces a node at a greater depth than its parents. Record the root-premise handles, rule identities and contents, relevant versions, and interpretation. Multiple descendants may share the same node and its basis rather than duplicate the acquisition history.

**Proposition 7D — Transfer of a selected, rooted derivation**

**Conditional proof-composition argument.** A first-discovery proof graph formed from (22b) is acyclic when edges run from premises to their derived conclusions. If its factual roots remain supported, its hypothetical roots remain explicitly assumed where used, and its rule steps retain an applicable truth-preserving interpretation in the new account, its conclusion transfers with those same conditions.

**Proof.** Every edge strictly increases discovery depth, so a directed cycle is impossible. Order the finite nodes by depth. The roots have their declared present basis. At each later node, the already supported parent conclusions and the applicable rule justify the node. Induction establishes the terminal conclusion, retaining any hypothetical conditions. ■

The first claim concerns the recorded proof, not the entire rule graph, which may contain cycles. An empty-body rule node still records the rule basis. The second claim is conditional, not an assertion that the system can certify all its roots or checker rules without assumptions. A root's label and a cryptographic hash identify part of the account; neither makes the root true.

The companion uses exact signatures as a sufficient reuse test within its fixed fixture contract. It rejects a changed premise, used-rule version, used-rule body, or interpretation; changing an unrelated rule does not invalidate the selected proof. Its historical proof object remains unchanged. If a different surviving rule path can establish the same conclusion, recomputation can recover it. Thus a failed applicability test means “this retained proof is not admitted by this test,” not “the conclusion is false.”

Only one selected proof per fact is retained. The implementation does not claim to enumerate every possible justification or perform optimal incremental invalidation. Exact matching can reject semantically valid transfer, and a

newly established interpretation may allow more than the test recognizes. This restricted check demonstrates an economical, inspectable seam; it does not settle the general cost of semantic applicability.

This is the bounded answer to the recursive question. A retained proof can point to what supports its therefore without being required to re-earn that entire history at every use. When a relevant basis changes, reopen that basis and its affected use. The account may stop at a declared floor without claiming that no further question about the floor can exist.

## 11.6 Charge shared work once, and keep the units visible

The propagation account can separate seed acquisition, construction or retrieval of the rule basis, present applicability checks, actual rule applications, dependency recording, and later retrieval. Use a sum only when its categories partition the charged work and use common units. Otherwise retain a profile—for example time, storage, and source accesses as separate quantities.

For a stipulated common-unit example, let seed acquisition cost 12, obtaining the rule basis 8, applicability 2, propagation 3, and recording 1, giving first-derivation cost  $D=26$ . Let extra packaging cost  $B=10$ , and each subsequent reuse cost  $R=4$ . One use costs 26 fresh versus 36 packaged; two uses cost 52 fresh versus 40 with reuse. Conversely, with  $D=4$ ,  $B=6$ ,  $R=5$ , reuse loses for every  $m \geq 1$ , since the reuse-minus-fresh cost is  $m+5 > 0$ . These values are stipulated work units, not timings or claims about a model's reasoning.

A shared event is charged once under a declared accounting interval. In a diamond derivation, one acquisition priced at 12 supports two one-unit branches and a one-unit join: total 15. Charging the shared acquisition once on each branch would produce 27 and double-count the same event. Omitting it would undercount. A separate execution that actually repeats acquisition owes its own event; structural resemblance alone does not make two costs the same.

Provenance sharing and cost sharing are therefore related but not identical. The proof graph tells which support a chosen derivation uses; the event account tells which work was actually incurred. Neither a large fan-out nor a short receipt guarantees that propagation is cheap. The seed, the rules, the representation, and the checks must all remain in the bill.

## 11.7 The whole family has a bill

The indexed interface does not remove the cost distinctions that motivated FRONT. A frame-family application may owe acquisition, calibration, alignment to a reference state, construction or retrieval of the update and readout maps, dependency checks, calculation, storage, decoding, and later reopening. Shared source work is charged once per actual event; independent measurements have their own costs.

Proposition 5C can certify that finitely many readouts carry the family's whole row span without supplying a cheap way to find them. Proposition 5E minimizes linear side-trace rank, not sensor price or conditioning. On  $Q^2$ , either  $y$  or  $2y$  repairs the  $y$ -query beside an  $x$ -readout. Both have rank one; a declared acquisition price of one for the first and one hundred for the second is consistent with that equal rank. Those prices are stipulated, not measured. Algebra alone does not choose the cheaper instrument.

Exact cancellation also differs from small attenuation. The readout  $x + y$  kills the nonzero difference  $(1, -1)$ . The invertible map  $(x, y) \mapsto (10^{-6}x, y)$  kills no nonzero vector, even though one exact output component is small. Noise, numerical precision, and physical acquisition limits would need an additional model before small values could be treated as indistinguishable. A nonlinear energy comparison belongs to its own fiber relation, not automatically to a linear kernel.

At the source level, knowing the laws  $L_w$  does not usually recover all  $r_f L_w x$  from one insufficient reading. Observing again supplies data; calculating from a fully specified state is derivation; changing coordinates is re-expression; learning a previously uncomputed consequence is another kind of work. Keeping those operations separate prevents the cost of obtaining the next distinction from vanishing inside a compact matrix product.

## 11.8 The explanation has a basis too

Feynman’s magnet/rubber-band discussion is useful here as an explanatory test: does the analogy add a usable picture, an independently supported mechanism, a proved equivalence, or a new testable consequence? Returning to the starting claim without additional support does not earn new warrant. But cycles are not prohibited: §11.3 already distinguishes an unsupported self-starting proof from a supported rule cycle, and sound equivalences can be informative. [S10, supplied 21:04–21:36]

The rule basis and the readout basis therefore belong in the same retained account. A failed applicability check on an old decoder means that route is unsupported now; it does not mean the requested answer is false. A later supported alternative can recover the answer without changing what the historical receipt recorded.

## 11.9 A bounded correction operator and its bill

CORRECT(record, context, grant, budget) specifies the implementation contract. Its obligations are to detect a typed discrepancy; bind it to a source, interpretation, version and scope; preserve the event; locate recorded affected uses; propose a remedy; check permission where an effect is involved; apply the approved candidate in isolated state; revalidate; and return a supported result or an explicit remainder. These are logical obligations, not nine compulsory model calls. Cheap operations may be fused without losing their receipts.

A selected proof is a finite rooted derivation DAG, not merely a reachability path. All premises of a rule are conjunctive; separate derivations of a conclusion are alternatives. Cyclic operational graphs may be recorded, but an unsupported cycle is not a grounded proof. For a fixed query  $q$ , the finite recorded set and its applicability predicate define:

$$\text{RecordedSupport}_t(q) \Leftrightarrow \exists d \in D_{rec}(q) \text{Valid}_t(d). \quad (\text{C1})$$

Valid checks the recorded roots and every rule, including required source/version/interpretation/scope witnesses. This is a definition relative to a finite candidate set, not completeness for truth or all possible derivations. A root challenge can invalidate one proof without invalidating an independent proof of the same proposition. Historical USED edges are retained separately from the current rule graph so removal of a link does not erase the record of earlier reliance.

### Proposition 7E — Conditional support and history preservation under correction

**Elementary proof-composition and transition invariant.** Fix a finite context, a declared interpretation  $h$ , and the recorded candidate derivations. Assume (i) accepted root validators support facts true in  $h$ ; (ii) each admitted rule preserves truth in  $h$ ; (iii) every returned affirmative witness is a grounded finite derivation whose conjunctive parents and current basis are checked; (iv) an effectful candidate is committed only under the required current grant and a successful declared postcondition; and (v) each transition appends history without altering its earlier prefix. Then every returned supported conclusion is true in  $h$  under that basis. Invalidating one selected derivation does not invalidate another derivation with an unchanged valid basis. Every committed modeled change satisfies the declared grant and postcondition, and the history prefix survives both successful and failed attempts.

*Proof.* For an affirmative witness, order its finite DAG topologically. Root truth follows from (i). At each rule node all parent facts are true by induction, so (ii) gives the conclusion. This proves the terminal query under its recorded basis. Disqualifying another derivation changes none of the checked nodes of an independent valid witness. By (iv), commit can occur only after both guards hold. By (v), each transition preserves the prefix; induction over finitely many transitions preserves the initial history, including failures. ■

These are conditional invariants, not an independent formal verification of the Python implementation or the world supplying its inputs. Freshness is fixed during the modeled call. Concurrent grant revocation, external changes between check and use, crash recovery, incomplete real-world dependency capture, and dishonest validators remain outside this result. The implementation’s executable checks test bounded instances of these obligations. No termination, completeness, cheapest-repair, or universal reversibility theorem is added.

The result vocabulary distinguishes supported-with-basis, conditional-on-roots, proposed-not-applied, revalidated-after-change, unresolved/no-valid-return-found, not-authorized, error, and quiet STOP. These operational types do not replace PAL claim statuses. The present executable lanes implement finite support/unresolved/stop, explicit grant tokens, and the scratch-repair outcomes. The broader conditional and open-ended evidence-acquisition interfaces remain proposed, not implemented merely because they have names.

The correction bill extends rather than replaces §11’s reuse accounting:

$$C_{correction} = C_{detect} + C_{bind} + C_{trace} + C_{altcheck} + C_{plan} + C_{grant} + C_{apply} + C_{revalidate} + C_{record}. \quad (C2)$$

Equation (C2) is an accounting schema. Shared reads are charged once; absent phases cost zero, not an invented measurement. The study records root/rule/premise checks, graph indexing and visits, closure passes, witness/history bytes, call wall/CPU time, input parsing, and separate oracle/logging costs. Schema categories are not claimed to equal independent hardware timers. In particular, a support-only run does not include a real file repair, a human reviewer’s labor, or model tokens. A finite operation-budget wrapper returns unresolved on exhaustion before issuing an incomplete witness; it is not a wall-time or memory sandbox. [S16]

## 12. P versus NP: exactly what this buys, and what it does not

For a language  $L$ , membership in NP means there are a polynomial bound  $p$  and a polynomial-time verifier  $V$  such that

$$x \in L \Leftrightarrow \exists w, |w| \leq p(|x|), V(x, w) = 1. \quad (23)$$

P asks for a deterministic polynomial-time decision procedure. SAT is NP-complete, so a uniform polynomial-time SAT decision procedure would establish  $P = NP$ . These are standard definitions, not FRONT’s restatement of the problem in different metaphysical terms. The Clay Mathematics Institute lists the problem as unresolved at the research date of 4 October 2026. [R1; R11]

### 12.1 Three bills that must not be merged

A proof may **exist**; checking a supplied proof may be **efficient in its length**; finding a proof may still require substantial work. A succinct description of an algorithm does not bound its running time, and a compact record of an earlier answer does not bound the first computation that produced it.

For UNSAT, universally polynomial-size certificates with a polynomial-time checker would give UNSAT membership in NP and hence  $NP = \text{coNP}$ . That conclusion is not itself a proof that  $P = NP$ . The Cook–Reckhow framework makes the corresponding equivalence precise for polynomially bounded propositional proof systems. [R2, §1; R3]

A cached proof can be long. A proof-checking procedure polynomial in the proof’s length can consequently take exponential time in the original input size when the supplied proof is exponentially long. Calling the receipt “checkable” does not remove its length from the account.

### 12.2 Nonunique next cuts do not prove hardness

A sorted-list lookup can have several harmless implementation choices. A graph search can choose among many frontier vertices and still run in polynomial time. Even SAT instances can have abundant witnesses that are easy to find. The mere existence of branching, several acceptable next steps, or a residue in a particular agent’s knowledge proves no lower bound.

Nor does  $P = NP$  require an algorithm to know an optimal next action in every environment. It would supply polynomial decision procedures for NP problems. It would not supply omitted observations, settle every undecidable problem, determine moral goals, or erase all future uncertainty. An eventual proof either way would leave FRONT's scope and provenance obligations meaningful.

### 12.3 A conditional accounting lemma

Fix one uniform deterministic algorithm on explicit SAT inputs, including its cache-building code. Let  $n$  be the encoded input length. Assume initialization costs at most a polynomial  $p_0(n)$ . Assume every complete macro-step—including selecting what to do, retrieval, source checks, proof handling, logging, and all administrative work—costs at most a polynomial  $p_2(n)$ . Assume there is a nonnegative integer rank throughout the run, initially at most  $p_1(n)$ , decreasing on every nonterminal macro-step. No instance-dependent advice or precompiled answer table is supplied. All auxiliary data must be constructed by this uniform algorithm within the charged bounds, measured in the original formula length  $n$ . Finally, assume each terminal return is a sound YES or NO and every input terminates; abstention is not a decision.

### Proposition 8 — Polynomial accounting consequence

**Conditional bookkeeping lemma.** Under those premises the algorithm decides SAT in polynomial time, within

$$p_0(n) + (p_1(n) + 1)p_2(n) \quad (24)$$

up to a fixed implementation factor. Therefore those premises for an actual SAT algorithm would imply  $P = NP$ .

**Proof.** There are at most  $p_1(n)$  nonterminal steps and one final step. Add their bounded costs to initialization. The assumed correctness and termination make the result a SAT decider. ■

This does not solve the hard part. It states its unpaid obligations. None of the reuse or fiber lemmas supplies the rank, the per-step bound, or the correct terminal decision for arbitrary SAT instances. Administrative returns hidden between macro-steps must also be bounded. A heartbeat is not a rank decrease. [S1, A13]

Conversely, a polynomial-time SAT decider could be instrumented to retain a polynomial-size transcript of its computation. Thus FRONT is not a complexity class that excludes efficient ordinary algorithms. Its accounting language becomes a substantive complexity result only when the missing uniform bounds are independently proved.

### 12.4 Restricted proofs and compilation

For resolution-based clause-learning solvers, restricted proof complexity supplies real limitations as well as positive results. Atserias, Fichte, and Thurley analyze a specific randomized restart-and-learning algorithm and its relation to bounded-width resolution; their bounds depend on the width parameter. Such results do not cover every possible SAT algorithm, and holding a parameter fixed is not the same as allowing it to grow with the input. [R9]

Knowledge compilation provides another direct comparison: pay an offline cost for a representation supporting specified efficient online queries. Its central trade-off involves representation size and supported operations. “Fast after compilation” therefore does not imply “fast including compilation.” [R7]

A table containing answers for every possible input length does not solve the uniform problem unless constructing, storing, and accessing the required information is accounted for. A finite physical resource limit is likewise not an asymptotic lower-bound proof.

## 12.5 What the finite zipper does not hide

Proposition 7A bounds strict rounds in an explicitly supplied finite vocabulary. It does not show that this vocabulary or its rules are small in an original SAT encoding, cheaply discoverable, semantically adequate, or complete for the requested decision. The unit-propagation example itself stalls without a supplied learned consequence. No step in the new model discharges Proposition 8's uniform construction, totality, or complete-decision obligations.

The useful conclusion is narrower: once a rule account and seed are actually supplied under the stated conditions, its successive consequences and their cost categories can be inspected. Any proposed general solver must also pay for reaching that situation. The recursion of explanations does not by itself establish a runtime lower bound, just as finite stabilization does not by itself establish an efficient general SAT decider.

## 12.6 Linear recovery is not an unpriced general decider

A finite-dimensional kernel calculation and a small side-trace rank do not establish that an arbitrary task has a small, cheaply constructed linear representation. Nor does a finite sufficient subfamily identify which probes to choose under a black-box interface. For a complexity claim, the original input length, coefficient encoding, map construction, observation acquisition, arithmetic, storage, and terminal decision obligations remain in the same bill. The new results refine the reuse contract; they do not discharge Proposition 8's general premises.

# 13. Neighboring work: compare the mechanism, not only the vocabulary

FRONT's components have substantial antecedents. The comparison below concerns reusable objects, validity conditions and costs; it does not claim that a new name establishes priority. References [R1–R13] retain the v0.3 source-inspection history. New comparisons [R14–R21] were checked for this revision with the access limits recorded in Appendix C. The expanded seven-axis map is supplied in `LITERATURE_COMPARISON_v0.4.md`.

| Neighbor                                    | Reusable object and validity route                                                | FRONT-facing question                                                    |
|---------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Memoization / dynamic programming           | Subproblem answer under an adequate key; recurrence and storage have costs        | What must the key preserve when the job changes?                         |
| Truth / assumption maintenance [R5; R6]     | Justifications and assumption environments; revise support as dependencies change | Retain historical warrant without treating it as present applicability   |
| Clause learning / incremental SAT [R4; R16] | Learned constraints and solver state; preserve assumption dependence              | The SAT laboratory is an instance, not a new solver                      |
| Knowledge compilation [R7]                  | Compiled representation; supported query and transformation families              | Query-relative serviceability is already an established concern          |
| Build Systems à la Carte [R14]              | Verifying / constructive traces; dependency hashes and early cutoff               | Which checking and rebuilding costs belong to this use?                  |
| Self-adjusting computation [R15; R21]       | Dependency trace plus memoization; change propagation                             | Compare the result with from-scratch execution and charge trace overhead |
| Provenance sketches / IMP [R17]             | Sketch of relevant data; cross-query applicability and update maintenance         | A provenance object can itself become stale or require repair            |

| Neighbor                     | Reusable object and validity route                                            | FRONT-facing question                                                            |
|------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Invalidation Contracts [R18] | Recovery suggestions with freshness metadata; validity and planner compliance | A valid reusable record and a model actually using it are different achievements |
| PACMI [R19]                  | Memory nodes and typed dependencies; cascading validity updates               | Preserve obsolete material as history without using it as current premise        |
| Agent Zero Memory [R20]      | Source-linked memory and evidence pointers; structured retrieval              | Inspectable source access must still support the claim attributed to it          |

Build-system traces are the closest mature engineering comparison to the exact-dependency lane. Self-adjusting computation offers a stronger established semantics of consistency with recomputation. Incremental SAT has specialized reuse far beyond the deliberately basic solver used in §16.1. FRONT supplies no head-to-head result against these systems. [R14–R16; R21]

**A correction to the planning key:** changing the query is not, by itself, a distinct FRONT invention. IMP explicitly considers whether a sketch captured for one query can serve another, then maintains stale sketches before reuse. Knowledge compilation likewise evaluates which queries a retained representation supports. FRONT should position its integrated warrant/continuation/cost account alongside these, not declare their questions absent. [R7; R17, §2]

The 2026 agent-memory papers are preprints, not independent validation of FRONT. Invalidation Contracts’ empirical setting uses synthetic APIs and distinguishes contract validity from planner compliance. PACMI’s indexed primary abstract supports its typed-dependency and history-preservation comparison; its full manuscript was not successfully accessed in this build, so no detailed theorem or benchmark claim is imported. Agent Zero Memory supplies a relevant provenance-aware retrieval design; its interface restriction on citations should not be promoted here into a general guarantee of semantic entailment. An opened source can still be misinterpreted. [R18, §§3.6, 6.4; R19, abstract; R20, §3.1]

The proposed FRONT contribution is the joint account of **historical warrant, current applicability, task-relative continuation, visibility/recovery and full cost**. Whether this integration establishes a distinct research contribution remains a question for comparison and external review. Useful bounded mathematics and an honest experiment can stand without a broad priority claim.

### 13.1 What the correction addition inherits

Truth-maintenance systems already separate beliefs from the justifications that support them. De Kleer’s assumption-based account represents conjunctive assumption environments, alternative support, and inconsistency information. FRONT does not claim to invent dependency propagation, independent-support rescue, or nogood recording. Its finite prototype is deliberately much smaller than a mature ATMS. [R5–R6]

AGM theory distinguishes contraction and revision and constrains choices among retained beliefs. Its selection machinery is an important ancestor of the question “which repair?” FRONT’s practical permission boundary adds a different issue: a mathematically admissible revision is not itself authorization to change an external artifact or disclose evidence. This distinction is not presented as a refutation or generalization theorem about AGM. [R22]

Consistent query answering studies answers that survive the relevant database repairs; the cited minimal-repair account quantifies across repairs. FRONT’s C1 instead asks whether at least one currently valid *recorded derivation* supports a fixed query. “True in every admissible repair” and “has one checked recorded proof here” must not be substituted for one another. [R23]

Provenance semirings make the AND/OR distinction concrete: joint contributions combine multiplicatively and alternative derivations add. That is closer ancestry for support multiplicity than an ordinary untyped edge list. Our chosen-witness prototype records one successful DAG rather than enumerating all semiring expressions or all consistent repairs. [R24]

FRONT integrates task-relative serviceability, preserved historical use, selective requalification, explicit effect grants, priced correction, and continuation/return. Its mechanism and finite artifacts can be inspected. Novelty of the integration, practical superiority, and completeness for external consequences are not established by the existence of a working prototype. The current comparison cannot stand in for a competitive incremental-solver or ATMS benchmark.

## 14. Why the spherical and Riemann branches belong nearby

Pang’s spherical-closure work asks what a support can carry when additional properties are demanded. Its relevant distinction is between scalar labeling or tiling and a globally continuous, nowhere-zero tangent vector field. The second requirement encounters the classical hairy-ball obstruction on  $S^2$ ; a mere demand for congruent surface pieces does not. [S6, §§2–5]

For the unit sphere, the explicit rotation field

$$v(x, y, z) = (-y, x, 0) \quad (25)$$

is tangent because  $(x, y, z) \cdot v = 0$ . It vanishes at the two axis poles. Congruent pole-to-pole lunes can still be carried into one another by rotations. Successful tiling and nonvanishing tangent motion are different claims.

The analogy to FRONT is **property-specific serviceability**: one representation may carry a present answer while failing a requested family of continuations. No topological theorem has thereby been transferred to computation. A seam in a diagram is not evidence of an exponential running-time bound, and FRONT imports no minimum physical area, horizon-entropy law, or quantum model of thought.

The Riemann–Pang Ledger provides a second neighboring lesson. Its finite arithmetic and certification work does not discharge its named all-tail obligations merely by extending a prefix. The document explicitly retains `O-ALL-EVENT-BOUND` and related transport, coverage, and stability obligations. FRONT uses this as an example of keeping a remaining proof obligation visible after useful local work—not as a derivation of any SAT result. [S5, p. 22]

An archival detail is worth preserving: the accessed RPL manuscript bears a pending-adoption cover, while a separate 12 August 2026 receipt records Pang’s adoption of a particular manifest. These are distinct source objects. Their presence does not justify treating the manuscript’s open mathematical obligations as solved. [S5a]

## 15. Executable evidence: families first, counts second

The retained companions exercise small, fully described families: Boolean SAT examples, finite-rule closure, source/basis changes, indexed linear readouts, decoder constructions and cost controls. Their many assertions are repeated checks over those bounded families, not thousands of independent experiments.

| Preserved component              | Family and boundary                                                                                                           | Check invocations |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|
| FRONT v0.2 legacy reconstruction | Resolution/staleness examples; 256 two-variable CNFs; 64 six-bit states; 343 common-action families; other specified controls | 2,422             |
| FRONT v0.2 zipper                | Six-variable chain, broken-link control and recorded waves                                                                    | 12                |

| Preserved component             | Family and boundary                                                                                 | Check invocations |
|---------------------------------|-----------------------------------------------------------------------------------------------------|-------------------|
| FRONT v0.2 positive-rule family | All 256 rule sets on two facts; 1,024 initial closures and 2,048 seed comparisons                   | 15,744            |
| FRONT v0.2 dependencies/bases   | Changed domains, used/unused rules, hypotheses, grants and cycles                                   | 24                |
| FRONT v0.2 cost controls        | Stipulated work units and shared-cost cases, not timing results                                     | 23                |
| Synthesis key v0.1              | Delayed visibility, return, invariance and 64 rational matrix/readout pairs                         | 287               |
| Synthesis key v0.2              | Indexed frames, restricted pairs, label loss, coordinate changes, 256 triples and 1,024 query cases | 2,862             |
| FRONT v0.3 integration          | State and residual query repair, typed sequences, quotient updates and counterexamples              | 1,724             |

The first five rows are one 18,225-check program. The other programs retain separate identities. Repeating them in normal mode, optimization mode or an extracted archive does not multiply the number of independent results.

## 15.1 The exact finite families

For the two-fact rule family, a rule chooses one of four premise subsets and either conclusion: eight possible rules, hence 256 rule sets. The checker considers all four initial fact sets and both seeds, compares closure with Boolean valuation enumeration, and checks stabilization, monotonicity, first-discovery depths and the wake decomposition. The maximum observed number of productive rounds is two, within this family only.

The linear family uses all 16 binary-entry two-by-two readouts over  $Q$ , four scalar queries and five generated subspaces per readout. This gives 64 state-query cases and 80 BRIDGE instances. Among 320 residual-query candidates, 160 annihilate the generated subspace and are in scope; the other 160 are not silently admitted. The code constructs side traces and exact decoders, and compares two-column ranks with a separate minor-based calculation.

The v0.3 groups contain 16 rank-oracle comparisons, 384 state-query repair checks, 480 typed-sequence checks, 800 residual-query repair checks, three common-baseline controls, four partial/restricted-domain controls, 26 update/return checks, seven dependency/acquisition checks and four single-added-view checks. The common-baseline control preserves  $x - y$  through the joint quotient while independent coordinate quotients erase it. These are code-level checks of the stated small constructions; manuscript arguments retain the general hypotheses.

## 15.2 Historical evidence remains historical

The v0.1 manuscript reports 2,402 assertions, but its original executable, result JSON and `ORIGIN.md` remain unrecovered in the supplied lineage. The available v0.2 companion is an explicitly labeled reconstruction, not a recovery of those bytes. Its 2,422 legacy-specification checks are not relabeled as the old 2,402.

That reconstruction retains the three-model SAT example, 64 applicable formulas among 256, 1,953 pairs merged by a six-bit OR summary, and the 343 action families. Its repeated-core illustration compares 512 assignment probes with 32 probes plus 15 proof rechecks and 30 premise tests. A competent solver can detect the contradictory units directly; this is transparent accounting, not competitive SAT performance. [S11]

The earlier source-inspection and render reports are preserved under `sources/FRONT_v0.3/`. Their dates and claims are not rewritten as fresh v0.4 work. No public FRONT v0.1 erratum target was established by the supplied materials, so none was edited or invented.

## 15.3 Fresh v0.4 verification

`front_checks_v0_4.py` imports the unchanged v0.3 runner and reruns its four components: **18,225; 287; 2,862; and 1,724**, separately. Its additional **688 focused audits** cover the new policy behavior, exact time partitions, source bounds, unavailable model metrics, certificate rejection, new-query handling, completion, reopening, conservative misses, an independently evaluated 16-formula unit-clause family, and source preservation.

These audits are separate from FRONT-REUSE-01's measured event log. They may execute the same fixtures to check code; they are not additional experimental replications. The runner uses explicit failures, suppresses bytecode writes into the ancestry and refuses an existing output basename.

```
python front_checks_v0_4.py --output checks_results_fresh.json
python -O front_checks_v0_4.py --output checks_results_fresh_opt.json
```

Fresh outputs record exact script hashes, environment and execution time. The complete build note records normal/optimized agreement, the final extracted-package check and visual inspection. These executable checks remain separate from the Lean companion described in §17.2. They do not establish independent human proof review or a model benchmark.

## 15.4 A separate measurement lane

FRONT-REUSE-01 D is the deterministic algorithm study in §16.1. Its 27 case pairs are frozen before execution; five timing repetitions do not turn them into 135 independent case pairs. Lane M is the distinct real-language-model protocol and remains **PREPARED\_NOT\_RUN**. There is no token-savings estimate to infer from the deterministic operation counts.

## 15.5 What was preserved and what was rerun

The original v0.4 source archive was checked against all 291 manifest entries before editing. Its FRONT-REUSE-01 fixtures and primary result files are retained unchanged and are not counted as new correction outcomes. The original v0.4 checker was rerun against a complete isolated source copy in normal and optimized Python; its new logs are retained separately from the source evidence. The full original archive remains the complete reconstruction context; the review ZIP includes a privacy-minimized legacy-code subset with a separately identified replay adapter. [S16]

The correction lane is a new retained reconstruction after an interrupted build. Temporary files from that attempt were unavailable, while the author's source artifacts remained intact. Earlier conversational result and timing reports are not presented as preserved experiment artifacts. The retained design, implementation, fixtures and protocol received a new pre-execution hash freeze. The normal and python -O runs agree on substantive outcome counts; timing varies and is not required to match. That within-build freeze is not external preregistration or blind design. [S16]

# 16. Branchline application: keep the support smaller than the work

A live Branchline study need not build a new controller. It can attach a scoped record and retrieval route to an existing event log. The synthetic study below is a separate preliminary lane, not a modification of that application. The natural starting workload is a sequence of related tasks with known outcomes: repeated technical questions, changed source versions, corrected interfaces, or small puzzles with established controls.

The model should be free to make an ordinary supported move without proving that its whole world is ready. FRONT would help it reuse a corrected distinction when relevant, preserve the source of a tool result, and report an unresolved dependency without disguising it as a negative answer. A tool error should remain about the tool unless additional evidence connects it to the task's world. [S4]

A learned habit in weights and an addressable source receipt are also different carriers. A Dream may improve how a model responds, while the provenance required to justify a particular factual claim remains in the archive. The manuscript does not count a conversational improvement as a validated general reuse method. MIND and the Quiet Layer both distinguish contextual availability, later retrieval, and parameter adaptation. [S2; S4]

## 16.1 FRONT-REUSE-01: controlled repeat-and-change study

**Evidence status:** deterministic lane D executed; real-model lane M **PREPARED\_NOT\_RUN**. This section instantiates v0.3's proposed three-way comparison without claiming that a synthetic solver stands in for an LLM or for Chris's live Branchline usage. The protocol and expected-validity ledger were frozen within this build before primary execution. They were not independently preregistered. [S13]

### 16.1.1 Workload, access and validity

There are 27 before/after pairs: the following nine classes at three scales. Every pair starts with a fresh cache and contains a first derivation plus one later encounter. The source files are immutable synthetic snapshots, not copies of the private application or corpus.

| Mutation class               | Why the later encounter differs                              | Old exact answer applicable?      |
|------------------------------|--------------------------------------------------------------|-----------------------------------|
| Unchanged repeat             | Same SAT source and question                                 | Yes                               |
| Irrelevant source change     | Only an unused note changes                                  | Yes                               |
| Relevant dependency change   | One summand increases by seven                               | No                                |
| Invalidated positive witness | A new clause contradicts a previously forced assignment      | No                                |
| Surviving negative core      | Original contradictory units remain after strengthening      | Yes                               |
| New query / omitted detail   | A sum is followed by a maximum question                      | No; source may still help         |
| Finished task                | Completion state remains DONE; unrelated note changes        | Yes; answer remains STOP          |
| Source reopening             | New query uses a previously discovered unique-topic document | No; source locator remains useful |
| Changed bytes / same answer  | A numeric list is reversed; its sum is unchanged             | Yes, despite the byte change      |

SAT instances have 4, 9 or 14 variables. Numeric sources have 8, 256 or 8,192 values. The basic solver explicitly enumerates assignments, calculates a sum or maximum, or reads a completion status. It is intentionally not an optimized SAT solver. The final row is an additional, predeclared stress control for conservative applicability.

The frozen ledger distinguishes same-query reuse from source-only reuse and specifies the expected current answer and its reason. Constructed numeric answers are algebraically known; SAT answers are checked by a separate product-enumeration evaluator and direct clause tests. That evaluator is not called by any measured policy. It is a second code path, not independent human review.

### 16.1.2 Three policies, one base solver

**A — no reusable cache.** Read, parse and solve each encounter from the current snapshot.

**B — answer only.** Retain an answer under the exact query and scope, without source-version checking. A hit returns it immediately. A different question has a different key and is solved normally; the baseline is not deliberately made to reuse a sum as a maximum. This is an unvalidated cache, not a representative best-practice incremental system.

**C — source/certificate aware.** Retain the answer, actual source path, content digest, interpretation/scope and a SAT witness or the originally proved UNSAT clause set where relevant. On a candidate hit, read and hash the whole current source. Identical bytes under the fixed interpretation permit reuse. Changed SAT sources permit reuse only if the old witness still satisfies the formula or the old UNSAT core remains included under the same semantics and variable domain. Other changes require recomputation with the same base solver, using already-read bytes rather than opening the file twice. Non-SAT byte changes are conservatively rejected.

For the reopening case, the first solve discovers a topic by scanning 16 files. Only C keeps that earned locator; its later query opens the identified file and rechecks its topic. Unique topics and fixed paths are part of this fixture contract. This tests paid discovery followed by targeted reopening, not general repository navigation after arbitrary structural changes. All policies have the same source access. C receives no expected answer, secret dependency oracle or more capable solver.

### 16.1.3 Outcomes

The following counts are one logical pass across 27 follow-up queries. Every first derivation was correct in this constructed family; all five technical repetitions returned the same substantive outcomes.

| Policy                | Correct follow-ups | Answer reuses | False reuses | Valid opportunities not used |
|-----------------------|--------------------|---------------|--------------|------------------------------|
| A: No cache           | 27/27              | 0             | 0            | 15/15                        |
| B: Answer only        | 21/27              | 21            | 6            | 0/15                         |
| C: Source/certificate | 27/27              | 12            | 0            | 3/15                         |

There are 15 valid same-query opportunities. C uses 12 and misses three: one reordered-sum case per scale. Its misses are **20% of valid same-query opportunities**, not incorrect answers. B's six false reuses are the three changed-sum and three invalidated-witness cases: **6/21 cache hits**, or **6/27 follow-up queries**. A's 15 unexploited opportunities describe its no-cache design, not a faulty applicability test.

The new-query cases succeed in every arm. The source-reopening cases use one source open per C follow-up versus 16 for A and B. Finished-task decisions remain STOP in all arms; no actual actions are executed. C retains a source/certificate route for every answer, whereas B's 21 unvalidated hits have only the answer. A's fresh computations have current source support but do not retain it as a reusable cache record.

### 16.1.4 The bill, including the first use

Five technical repetitions use cyclic arm order and fresh per-pair caches. The entries below are **milliseconds for the complete 27-pair workload**, not per-request latency. Ranges are observed minima and maxima across the five runs; no confidence interval or population claim is made.

| Policy | First-turn median | Follow-up median (range) | First + follow-up median (range) |
|--------|-------------------|--------------------------|----------------------------------|
| A      | 54.183            | 56.146 (51.740–61.115)   | 113.523 (104.739–121.060)        |
| B      | 54.541            | 8.904 (7.731–9.624)      | 63.307 (59.738–64.187)           |

| Policy | First-turn median | Follow-up median (range) | First + follow-up median (range) |
|--------|-------------------|--------------------------|----------------------------------|
| C      | 55.289            | 20.258 (19.130–21.056)   | 74.419 (71.215–76.985)           |

A complete-run median is calculated from each repetition’s combined total, so it need not equal the sum of the two separate medians. Timing includes instrumentation, source reading, checking and serialization for the in-memory cache. It excludes evaluator work and synthetic-fixture creation. Operating-system caches were not flushed; these are warm-filesystem microbenchmarks on this container, not deployment latency or energy measurements.

The repeat-side exact operation counts are more portable than these very small times:

| Repeat-side category            | A       | B      | C       |
|---------------------------------|---------|--------|---------|
| Base solver invocations         | 27      | 6      | 15      |
| Full-assignment visits          | 67,648  | 0      | 16,912  |
| Arithmetic steps                | 33,818  | 16,906 | 33,818  |
| Source opens                    | 72      | 51     | 27      |
| Source bytes read               | 166,180 | 83,370 | 164,620 |
| Candidate applicability checks  | 0       | 0      | 21      |
| Hash bytes during applicability | 0       | 0      | 82,810  |
| Certificate steps               | 0       | 0      | 45      |

B’s zero repeat assignment visits partly reflect wrong cached answers, not free correct reasoning. C still reads almost as many source bytes as A: freshness was not supplied for free. Its three reopening cases remove repeated file discovery; its retained SAT certificates remove assignment enumeration. Arithmetic work is unchanged relative to A in this workload.

C’s median repeat-side pure checking time was 0.225 ms; source-read, lookup, derivation, packaging and other wrapper overhead are separate fields in the raw log. Source reading required for validation is **not** included in that pure-check number and is still included in total time. Canonical serialized cache sizes sum to 4,780 bytes for B and 10,540 for C across the separate pair-final caches; their maximum live per-pair sizes are 275 and 642 bytes. A retains no reusable cache. These are not Python heap or whole-archive measurements. No persistent-cache disk writing is timed.

### 16.1.5 What the result does and does not settle

C reduces repeated assignment work without any false reuse on this family, but the family was constructed to exercise specific distinctions. Its aggregate timing advantage is driven largely by the intentionally basic SAT enumeration and the locator exercise. It is not evidence of superiority over incremental SAT, build traces, or another efficient dependency-aware cache.

The changed-bytes/same-sum controls expose a genuine limit of the chosen check. C recomputes all three still-valid answers and is slower than A in their median follow-up timings. The large case takes 1.587 ms versus 1.392 ms in the primary run. Detecting change was cheaper than re-solving, but **detecting that the change was harmless was not supplied by the hash**. This is precisely the unresolved semantic-applicability problem, not a reason to erase the misses.

All LLM token/call fields are null. No suitable pinned model runtime or weights were supplied for this experiment; conventional local runtime/cache checks found none available, and no paid or external model call was made. Lane

M’s protocol fixes the conditions, source-access rules, required token/timing receipts, five paired repetitions, and treatment of initial model failures; model identity and budget must be fixed before that separate execution. **Measured model/token savings remain open.**

To reproduce D from a new output directory:

```
cd experiment
python run_reuse.py --output results_reproduction
```

The runner verifies the frozen inputs before and after, records its own hash and environment, and preserves all 810 measured events (27 pairs  $\times$  two encounters  $\times$  three policies  $\times$  five technical repetitions). Reproduction tests repeat the implementation; they do not increase the independent workload size. [S13]

## 16.2 Stopping is one of the outcomes

A task can finish. A user can withdraw it. An evidence path can become unavailable. A finite budget can end. These are different reasons to stop and should leave different records. None requires a performance of disappointment or another indefinite readiness stage.

The trace should make the stopping point intelligible to a later run. It should not obligate that run to continue without a current task or grant.

## 16.3 Key Turn: a reason-bearing handoff between episodes

Pang describes a practical division of work: a “Key turn” prompt states the intended outcome, destination, constraints, and details; a thinking episode discusses rough choices and prepares a workflow; a build episode works from that handoff. His report of faster and smoother builds motivates this illustration. It is not a controlled measurement, and the conversational “10x” is not treated as an experimental result. [S8]

The retained object should carry reasons, not only commands. “Do X to preserve Y” gives execution a basis for finding another way to preserve Y when X fails. The build episode still reasons about actual sources, code, tools, and newly encountered conditions. It need not repeatedly reconstruct the assignment or reopen unchanged decisions merely because it is a new episode. Conversely, a plan does not certify its own feasibility, extend a grant, or prove that implementation has succeeded. This is a small application of scoped resurgence and continuation sufficiency, not a new compulsory planning ritual.

## 16.4 A handoff can retain the labels that make reuse honest

The same result text can support different continuations when its source, version, or chair attribution differs. A reason-bearing handoff should therefore retain the relevant labels or a route back to them, not just the conclusion. Equation (12b)’s indexed-family example is a small mathematical analogue: removing labels can merge distinguishable states even when all numeric values survive. It is not a measured model of an agent.

An execution episode can reuse a selected answer whose basis remains applicable and reopen the implicated readout or dependency when it changes. It need not reconstruct every original state or inspect every archived turn first. The required carrier is determined by the protected question and current grant, not by a demand to make all of the past active at once. A key can remain available while the user chooses not to build; continuity need not become a summons.

## 16.5 FRONT-CORRECT-01: the refused premise remains consequential

The frozen retained workload contains **75 fixtures**: 25 templates in 16 adversarial families, at chain lengths 4, 16 and 64. Three completed tasks ask no query; the remaining cases ask q, z and an unaffected safe conclusion, giving **216 support decisions per arm**. P supplies a challenged root; a conjunction of A and B can provide independent support. Source versions, permission scopes, historical uses and current rules are explicit fixture data. This is neither a trained model nor a live Branchline/Pony Post deployment. [S16]

**A, unchecked continuation**, keeps the prior answer and proof without validation. **B, blanket invalidation**, reopens recorded downstream entries and runs a basic whole-graph Horn closure separately for each requested invalidated conclusion. **C, selective witnessed correction**, checks recorded alternatives on demand, validates conjunctive premises and current basis, and retains successful proof subgraphs. Cycles cannot support themselves. B deliberately lacks shared-query recomputation and other mature optimizations; speed relative to B must be read with that limitation.

| Measure                           | A   | B   | C   |
|-----------------------------------|-----|-----|-----|
| Correct support decisions /216    | 132 | 216 | 216 |
| Invalid affirmative warrants      | 114 | 0   | 0   |
| False positive decisions          | 84  | 0   | 0   |
| Valid affirmative witnesses       | 102 | 132 | 132 |
| Honest unresolved                 | 0   | 84  | 84  |
| Whole-graph recomputations        | 0   | 144 | 0   |
| Unnecessary blanket invalidations | 0   | 60  | 0   |

A's 114 invalid affirmative warrants include 30 cases where the conclusion has another current justification. Correct answer bytes do not repair the particular proof returned. B and C each requalify 30 formerly invalid selected routes: 24 through independent alternatives and 6 through a changed root version. Those are separate mechanisms, not one inflated "independent discovery" counter.

All three arms preserve the 75 supplied history prefixes. All share the same explicit grant gate: 21 requests, 3 permitted tokens, 18 denials and no executed external actions. Zero grant violations in this finite check is therefore **not an advantage specific to C**, and no real sandbox or authentication claim follows. Diagnostic types and trust metadata are supplied by the fixture; the program preserves their distinctions, rather than inferring intent or real-world authority.

One warmup precedes five technical repeats in rotating arm order, totaling 1,125 measured policy calls. Median call time including fixture read/parse was A 0.138 ms, B 0.304 ms, and C 0.224 ms on this container. Comparing per-fixture medians, C was faster than B in **67/75** cases and not faster in **8/75**. Raw events retain every loss, operations, phase timings and provenance sizes. These are warm-cache exploratory microbenchmarks, not statistically general estimates or a claim against optimized truth-maintenance systems. Shared host load, Python allocation, input/output and weak-baseline choices limit interpretation.

The separate **96-case scratch-repair assay** tests every combination of two endpoint charges, an optional corridor hole, an orientation lock, a mutation grant, and a candidate orientation. It returned 16 unchanged-supported outcomes, 24 denials, 8 revalidated changes and 48 no-valid-return outcomes. **16** granted candidate changes failed a secondary condition and were rolled back in isolated dictionary state; their failed attempts remain in history. The independent code-path oracle constructs a binary row and counts internal zeros rather than reusing the repair validator's sum. This tests a finite transaction model, not irreversible real-world repair.

Supplemental checks reproduce all eight source OR counts across **5,040** orders, compare **1,024** two-atom Horn systems / **2,048** queries with explicit Boolean valuation tables, and execute **525** operation-budget cases. The budget wrapper admits no partial witness and returns unresolved on exhaustion; the results include supported, unsupported, exhausted and quiet outcomes. **8** deliberately corrupted-output controls are detected. They are intentional negative fixtures, not undisclosed failures in the clean implementation.

The oracle and evaluated implementation use different algorithms but were written by the same assistant. Independent review and real-model execution remain open. The Lean companion checks mathematical statements and bounded correction models; it does not establish that this Python implementation refines those models. [S17] The strictest supported conclusion is that the specified finite procedures, fixtures and conformance controls satisfy their tested contracts, with measured costs and all named limits attached.

## 17. Open work and conclusion

The deterministic study makes the applicability cost concrete but does not remove the main research problem. The following work remains:

**Economical semantic applicability.** Preserve justified reuse across harmless representational changes without making validation as expensive as recomputation. The reordered-sum controls show the gap between a cheap sufficient test and full recognition of valid transfer.

**Priced observation selection.** Proposition 5E identifies sufficient side-trace capacity, not an inexpensive way to find or acquire an observation. The admitted pair family must still match accessible evidence.

**Compact reopening.** A source locator can save discovery, but broader histories must preserve enough warrant without forcing whole-archive replay. Changed paths, multiple candidate sources and competing derivations need explicit contracts beyond the unique-topic fixture.

**Noisy or partial evidence.** Move beyond exact signatures and noiseless linear readouts with a declared error model. False reuse and missed reuse must remain separately visible; a confidence label alone is not a calibration result.

**Changed representations and large stores.** Study cross-representation transfer, index maintenance, eviction, acquisition and retention under many stored objects. First construction and shared infrastructure costs remain on the bill.

**Actual model and workflow behavior.** Lane M is still unrun. A model may ignore a valid record or produce an invalid first answer, so validity and successful use cannot be collapsed. A live Branchline or Key Turn study must charge preparation and execution, preserve model/version/configuration and failures, and compare matched workloads. A narrow literature comparison is now present; a broad priority claim and independent review are not.

Useful continuation families narrower than all possible future questions, grounded rule discovery, and resource-bounded decision sufficiency also remain open. These results do not discharge the general complexity obligations in §12 or change BRIDGE's Hodge scope.

FRONT's organizing statement is:

**Do not pay to rediscover an applicable answer merely because the next task arrived. Do not pretend that retaining the answer resolved the next task.**

Two further principles make the cost of propagation explicit:

**Do not confuse the cost of finding the next cut with the cost of propagating what that cut makes possible.**

**Do not confuse the cost of propagating a distinction with the cost of establishing that the propagation rule applies.**

The second sentence does not cancel the first. It brings the rule basis into the same account. The zipper can carry earned structure forward without claiming that its teeth aligned themselves.

**Carry earned structure without erasing the distinctions a later task may need. Across the views and continuations actually admitted, establish what remains recoverable and what justified reuse costs.**

The seam can remain without being worshipped, erased, or blamed. Sometimes it marks an input that must be obtained. Sometimes it marks computation still owed. Sometimes it marks a changed job. Sometimes several good roads remain and the traveler can simply choose one.

There is room for an earned YES, a hard NO, an honest unresolved return, and a completed local task. The work is to preserve which one happened, why it applied, and what it leaves possible.

*Reviewer 2's note: The frontier is allowed to remain. Unpriced work is not allowed to vanish into it. The residual may be exactly zero; the giggle is not a theorem.*

### 17.1 Scope and remaining work

FRONT v1.0 provides an inspectable finite correction mechanism and its bounded experiment, while retaining the applicability-cost question that motivated FRONT. It does not make correction universally cheap, automatic or complete. A settled NO, a surviving historical event, an unsupported selected proof, an independently requalified conclusion and an authorized repair remain distinct objects.

Independent mathematics/CS and authority-boundary review remains open. The Single-Cut example uses the supplied local v0.1 file; its match to the public record remains unverified. Real-model performance, concurrent grant freshness, crash-safe external repair, complete capture of real-world consequences and priority remain open. The Lean coverage and its remaining formalization limits are stated below. [S14–S17]

### 17.2 Lean companion and formalization limits

The Lean 4 companion covers all 21 headed propositions. Seventeen have general mathematical statements covering the named claims; Propositions 1, 7D, 7E and 8 have explicitly bounded formalizations or a checked numerical core. All statements retain their declared hypotheses. C1 remains a definition and C2 remains an accounting schema. [S17]

The checked inventory contains 81 new public theorem declarations and 43 inherited theorem declarations, together with definitions and constructions. These totals include helpers, counterexamples and wrappers and are not counts of discoveries. The 163-declaration signature and axiom inventory, build logs and 13 module kernel checks are retained. Inspected axiom dependencies are confined to Lean's `propext`, `Classical.choice` and `Quot.sound`; the exact subset is recorded for each declaration.

The remaining bridges concern a general representation of admissible calculi and assumption discharge (Proposition 1); a verified shared-DAG representation, parser and exact-match implementation (7D); general implementation refinement, real authority, concurrent revocation and crash recovery (7E); and the uniform SAT-machine and complexity interpretation beyond the numerical cost bound (8). No SAT algorithm satisfying the latter hypotheses or resolution of P versus NP is constructed.

The companion reuses identified APCI and BRIDGE proofs with their original namespaces and byte identities. Its formal checking is separate from experiment execution and from independent scientific review. The manuscript preserves the audited mathematical statements; an editorial concordance binds the final source bytes to the earlier proof receipts.

## Appendix A. Corrections retained from the originating dialogue

The following distinctions qualify the exploratory claims that motivated FRONT.

**“The next cut is not uniquely determined.”** That alone is not a complexity obstruction. Proposition 4 requires a common adequate continuation, not uniqueness.

**“There is always unresolved residue.”** The surrounding world may remain open, but a declared finite task can be completely decided and stopped. FRONT does not impose artificial uncertainty on a proved local result.

**“The hard part is knowing which distinction matters next.”** This can identify a useful research problem, but selecting a move, computing a consequence, obtaining missing input, and verifying a transfer are different tasks. No equivalence to P versus NP is asserted.

**“Every NO can be reused.”** Only an established negative under a valid present applicability basis can be reused as such. An old refusal remains a historical fact; it is neither a universal physical impossibility nor an eternally renewable permission context.

**“Keeping the trace is enough.”** Retention, retrieval, source interpretation, proof checking, and continuation preservation have separate costs. A bare hash is not a reconstruction route.

**“Closure has a price.”** The requested property determines the obligation. A topological obstruction is not a universal storage or computational lower bound without an explicit adapter and proof.

**“Repeated full closure describes propagation waves.”** A completed closure is idempotent. FRONT uses a one-step rule operator and defines closure as its eventual fixed point, leaving intermediate work visible.

**“The original learned NO already supplies the chain.”** In the original SAT example,  $\neg a$  leaves three models and does not fix  $b, c$  individually. Equation (9a) visibly supplies new clauses and fresh variables for the chain.

**“Everything exposed after the seed was made by the seed.”** Proposition 7C separates unfinished consequences already available from the initial account from the seed-dependent difference. Semantic entailment and accessibility to a restricted procedure also remain different.

**“The observations confirm the traveling-front model.”** The bacon and mirror reports motivated a resemblance; no spatial propagation, common mechanism, or relaxation oscillator was established. The assistant’s exploratory continuous equations are not physical findings of this revision.

**“Following ancestry resolves the final foundation.”** The new model supplies a bounded, conditional account of rule use. It neither reduces all foundations to permission or invention nor requires endless recursive justification before action. These are corrections to assistant extensions, not replacements for Pang’s recorded questions.

**“All views after all continuations are available.”** The index set must specify actual pairs; a contract or a mathematical product does not certify acquisition. The observed subset governs the present decoder.

**“A detected effect identifies its source.”** Equation (20g) supplies a counterexample. A new distinguishing view can repair a query, but a changed output alone need not select its cause.

**“Related views can all be generated from one view.”** Proposition 3B states the missing factorization condition. Invertible coordinate changes preserve information; dropping labels can lose it.

**“Every view makes the source fully known.”** The family  $x+a y$  never reveals  $z$ . A finite row basis can represent an entire family’s kernel without making that kernel zero or the probes cheap to obtain.

**“Quotient each view separately; the family is unchanged.”** Independent generated baselines can erase a cross-view relation that the joint quotient preserves. A quotient’s existence also does not establish compatible dynamics.

**“Small, zero, hidden, and unpromoted are the same.”** Exact cancellation, small invertible attenuation, a hidden difference, and an account’s certification status have different definitions. Pang’s “matters” correction concerned matter/mass; the task-relevance analysis is not retroactively attributed as his intended meaning.

**“The familiar analogy supplies its own foundation.”** An unsupported circular restatement adds no warrant. This does not invalidate a sound equivalence or a supported feedback loop. Feynman’s examples remain motivating source material. Pang placed his first viewing in his twenties, about fourteen years earlier, not as a fourteen-year-old.

These corrections preserve the development of the argument. They do not change the authorship of FRONT’s name or the conceptual observation that motivated it.

## v1.0 correction history

The new dialogue distinguished attempting a spend from admitting an arrangement; a Boolean false label from a lie; a proven rejection from a forced repair; and a false statement from the real history of relying on it. The implementation

further separates conjunctive proof parents from alternative supports, historical dependency from current edges, and independent-route rescue from a new source-version witness. The interrupted attempt's unrecovered measurements are not reused as retained results. No assistant episode is represented as an independent referee solely because it carries a different name. [S14–S16]

## Appendix B. Claim and evidence ledger

| Item                                         | Evidence status                                 | Basis and remaining obligation                                                               |
|----------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------|
| FRONT name and expansion                     | User-origin provenance                          | Current conversation and Pang's authorship statement [S0]                                    |
| Scoped reuse, Propositions 1–2               | Elementary manuscript proofs                    | Fixed logic, valid premises, explicit transfer scope                                         |
| Fiber criterion, Proposition 3               | Inherited/restated                              | CHARTER Corollary 10.4; no new priority claim                                                |
| Common-continuation criterion, Proposition 4 | Proposed formulation; elementary proof          | Supplied finite adequate-action sets; no efficient discovery claim                           |
| Future-output preservation, Proposition 5    | Standard argument restated                      | Declared transition system; online update and efficiency separate                            |
| Coordinate lower bound, Proposition 6        | Elementary counting proof                       | Fixed-length exact encoding, all coordinate queries, no external reread                      |
| Cost crossover, Proposition 7                | Exact algebra                                   | Declared equal units and cost assumptions                                                    |
| Polynomial consequence, Proposition 8        | Conditional                                     | Uniform algorithm, polynomial steps and work, correctness, totality still owed               |
| Finite stabilization, Proposition 7A         | Elementary manuscript proof; finite checks      | Explicit finite vocabulary and grounded rules; acquisition and representation costs separate |
| Conditional soundness, Proposition 7B        | Elementary manuscript proof                     | Truth of roots and soundness/applicability of rules are assumptions, not self-certified      |
| Exposed-wake decomposition, Proposition 7C   | Elementary manuscript proof; finite checks      | Fixed rule account; two closure computations not free causal attribution                     |
| Selected-proof transfer, Proposition 7D      | Conditional manuscript argument; fixed fixtures | Rooted chosen proof, preserved bases; sufficient matcher need not find every valid reuse     |
| Finite fixtures                              | New replacement companion executed              | 18,225 checks; original v0.1 code unavailable; no real-model benchmark                       |
| Agent benefits and Key Turn                  | Proposed experiment; user report                | No controlled deployment measurement in this revision                                        |
| Spherical/RPL connection                     | Interpretive source lineage                     | No transfer of topological, physical, or RH theorem to complexity                            |
| P versus NP or NP versus coNP resolution     | Not claimed                                     | No corresponding proof supplied                                                              |

| Item added in v0.3                      | Status                      | Basis and remaining boundary                                         |
|-----------------------------------------|-----------------------------|----------------------------------------------------------------------|
| Indexed family, Proposition 3A          | Elementary proof            | Supplied labeled maps; data acquisition is separate                  |
| View factorization, Proposition 3B      | Elementary proof            | Common domain; decoder on the reachable image                        |
| Combined interface, Proposition 5A      | Inherited criterion applied | Only the specified pairs; no free counterfactual observations        |
| Persistent invisibility, Proposition 5B | Standard argument, restated | Known linear update; full-domain criterion and finite dimension      |
| Finite subfamily, Proposition 5C        | Elementary proof            | Existence, not inexpensive probe selection                           |
| Typed endpoint, Proposition 5D          | BRIDGE theorem instantiated | Shared generated baseline; canonical subquotients                    |
| Side-trace repair, Proposition 5E       | BRIDGE argument applied     | Finite linear rank; capacity, access, and price distinct             |
| Quotient update, Proposition 5F         | Elementary proof            | Invariance required for a well-defined update                        |
| v0.3 integration fixtures               | Freshly executed            | 1,724 new checks; three unchanged programs rerun separately          |
| Feynman and room-wide views             | Attributed motivation       | No inference of a physics result or a measured perceptual capability |

## B.1 Additions and remaining obligations in v0.4

| Item                                           | Status                                   | Boundary                                                                |
|------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------|
| Applicability-check hinge and contribution map | Editorial integration                    | No new general applicability algorithm claimed                          |
| New literature comparison                      | Primary-source review with access limits | PACMI abstract only; priority remains open                              |
| FRONT-REUSE-01 D                               | Executed deterministic study             | 27 constructed pairs; five technical timing repeats                     |
| False / missed reuse                           | Recorded separately                      | B: six stale hits; C: three conservative misses                         |
| Full-cost measurement                          | Executed within fixture boundary         | Hashing, reads and first packaging charged; no heap/energy/LLM estimate |
| FRONT-REUSE-01 M                               | PREPARED_NOT_RUN                         | Requires pinned model, settings, budget and actual receipts             |
| Historical checks                              | Freshly rerun, originals preserved       | Counts remain separate bounded checks                                   |
| Live Branchline modification                   | Not performed                            | Requires a separate user request                                        |

**Correction results and evidence limits**

| Addition                          | Status and evidence                                              | Boundary                                                                                          |
|-----------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| C1 recorded support               | Definition on a finite recorded derivation set                   | Not all truths or all repairs                                                                     |
| Proposition 7E                    | Conditional argument in §11; bounded Lean model in the companion | Supplied validators, rules, grants and postconditions; no general implementation-refinement proof |
| C2 correction bill                | Accounting schema; recorded phase/operation measurements         | Not independent costs or a general speed theorem                                                  |
| FRONT-CORRECT-01                  | Executed retained reconstruction, normal and optimized runs      | Synthetic fixtures and basic baselines                                                            |
| Scratch correction                | 96 finite state-transition checks, separate row-count oracle     | No live filesystem or irreversible-effect recovery                                                |
| Source preservation               | Manifest and exact-heading/display comparison                    | Preservation is not independent proof review                                                      |
| Independent adversarial referee   | OPEN / not performed                                             | Same-assistant review is not a substitute                                                         |
| Public SCT artifact/version match | OPEN / retrieval unresolved                                      | Local v0.1 only is inspected                                                                      |
| Real-model study                  | PREPARED_NOT_RUN                                                 | No model-performance or token-saving result                                                       |
| Lean companion                    | 21 propositions mapped; 17 general and four bounded              | Exact assumptions and remaining bridges are recorded in §17.2 and [S17]                           |

## Appendix C. Sources, attribution, and verification notes

The source history preserves the supplied FRONT v0.3 baseline and its nested sources, including BRIDGE v0.4, byte-for-byte. The v0.3 source note records its earlier native-equation and render inspections. Those historical inspections and the source descriptions [S0–S11; R1–R13] below are retained as ancestry, not relabeled as fresh checks of every original source.

New research for v0.4 inspected the author/publisher sources in [R14–R21], including the relevant build-trace formulas and IMP diagram as page images. PACMI’s primary indexed abstract was accessible, while its full text was not; the comparison is limited accordingly. The complete access map and the execution/environment records are supplied beside the manuscript. No external research paper is redistributed. [S12; S13]

### C.1 Author corpus and originating material

[S0] **Christopher D. Pang and Hearthline-assisted drafting conversation.** FRONT origin, 2 October 2026 in the supplied local chronology. Version 0.1 reports that primary naming statements and conceptual corrections were preserved in its `ORIGIN.md`; that original file was not recovered in this build. The v0.1 manuscript remains the source for its naming account. No exact second-level timestamp or universal first-use claim is invented.

[S1] **Pang, Christopher D. *PAL v2.4: Mechanical Structural Spine*.** 22 September 2026. DOI [10.5281/zenodo.22888036](https://doi.org/10.5281/zenodo.22888036). Consulted supplied DOCX, especially §§1–4, the A9/A12–A15 roles, closure grammar,

and boundary/projection distinctions. Governs PAL terminology here. PAL’s mathematical realizations are not inferred wholesale from this spine.

[S2] Pang, Christopher D. *MIND: Matrix of Information, Negation, and Distinction*, v0.4. 21 September 2026, publication-candidate draft. DOI as printed: [10.5281/zenodo.22882309](https://doi.org/10.5281/zenodo.22882309). Consulted supplied PDF, especially §§1, 3–5, 8.7, 9 and its scope notes. Its original PAL v2.3 placement remains historical; FRONT uses v2.4 for PAL terms.

[S3] Pang, Christopher D. *CHARTER: Certification and History Architecture for Robust Transport, Evidence, and Retention — Framed Prime Shells and Bound Persistent Shells*, v1.0. 5 September 2026. DOI as printed: [10.5281/zenodo.22288471](https://doi.org/10.5281/zenodo.22288471). Consulted supplied 45-page PDF, especially §§10, 15, 16.3. The source distinguishes a limited inherited Lean core from later manuscript-level material; FRONT does not enlarge that machine-checked boundary. Public deposit was not independently audited here.

[S4] *The Quiet Layer: Information Without an Acknowledgment Demand*. Supplied conversation note, file `Pasted markdown (20260924-164758).md`. Conceptual and engineering antecedent from Pang’s collaboration with Hearthline; no journal status or independent neuroscience validation inferred. Consulted core definition, deferred relevance, temporal comprehension, ARC illustration, and limitations.

[S5] Pang, Christopher D. *The Riemann–Pang Ledger*, v1.0. 11 August 2026 manuscript. DOI as printed: [10.5281/zenodo.21894278](https://doi.org/10.5281/zenodo.21894278). Consulted status, abstract, claim classes, and p. 22 obligations; no analytic result is a premise of a FRONT proposition. Raw-byte materialization was unavailable for the accessed manuscript; the inspected text remains the basis of this citation.

[S5a] *Riemann–Pang Ledger v1.0 Author Adoption Receipt*. JSON, adoption time recorded as 12 August 2026, 00:51:40 UTC. Records exact-manifest adoption within its stated chat-account trust context. The receipt does not itself prove publication, mathematics, or an independent digital signature.

[S6] Pang, Chris. *The Price of Spherical Closure: A PAL-Based Formalization of Binary Pattern Obstruction on Spherical Layers*. Supplied 12-page public working paper, copy `The Price of Spherical Closure(6).pdf`. Consulted §§2–5 and limits; no date or DOI invented for this copy. Historical  $\Omega$ -shadow language is not adopted as current PAL definition. Used only for the property-specific closure analogy.

[S7] *A path in the middle*. 29 September 2026. Hearthline’s conversation-derived note, with Chris Pang’s signed addition; supplied file `2026-09-29-a-path-in-the-middle - Chris Signed.md`. Used for the separation between preserved ancestry and growing source-linked connections, not as evidence for an algorithmic theorem.

[S8] Pang, Christopher D., and Hearthline-assisted continuation. Selected v0.2 conversation passages and the production workflow, 2 October 2026 conversational date. Included as `sources/FRONT_v0.2_Conversation_Seeds.md` and `sources/FRONT_v0.2_Production_Workflow.md`. The supplement distinguishes verbatim user statements from assistant synthesis and is not a complete transcript or the original `ORIGIN.md`. It supplies the zipper observation, recursive rule question, Key Turn report, and requested revision direction—not independent scientific corroboration.

[S9] Pang, Christopher D. *BRIDGE: Binding Residue and Information Dynamics of Generative Export*, v0.4. Research draft dated 11 September 2026; printed DOI [10.5281/zenodo.22699551](https://doi.org/10.5281/zenodo.22699551). Supplied `BRIDGE_v0.4(1).docx`, preserved unchanged. Imported sections: “Frame First Geometry and Target Contract,” “Detection Binding and Account Promotion,” “Four Sector Endpoint Theorem,” “Frame Morphisms and Certificate Transport,” “Information Dynamics,” “Protected Answers and Minimal Repair,” “Monodromy Return,” and “Residue Register.” Relevant native equations were checked from the actual Word source and its fresh render; indexed extraction omitted many equations. The new render has 36 pages versus the older 32-page index, so section headings are the primary locators. No fresh public-deposit or full Hodge audit is claimed.

[S10] Feynman, Richard. *Fun to Imagine*, user-supplied transcript selections. The supplied timestamps identify the radio/readout example (42:50–43:13), the magnet/rubber-band explanation (21:04–21:36), and the “Ways of Thinking” discussion. Selected text and the two complete supplied later-section files remain in the key’s `supplements/` directory. These are supplied historical speech, not an independently verified audiovisual transcript or updated survey of physics. Only the explanatory motifs used in §§1.4 and 11.8 enter this paper.

[S11] **Pang, Christopher D., and Hearthline-assisted continuation and build keys.** 3–4 October 2026 supplied chronology; active target instruction `FRONT_v0.3_Build_Key.md`, previous synthesis keys v0.1–v0.2, source-labeled conversation supplements, and the request to build FRONT v0.3. The indexed-viewpoint idea, “all of it at once” correction, attribution correction, and selected destination originate with Pang. The selected exact adapter, implementation, and editorial placements were prepared with AI assistance. In the preserved v0.3 package, the complete input handoff is byte-preserved under `sources/build_key/`; historical activation prompts there do not supersede the active target.

[S12] **FRONT v0.4 planning and review input.** Chris supplied Claude’s second-review text in the conversation, requested research and a key, then explicitly turned the key. `FRONT_v0.4_Research_and_Build_Key.md` is preserved under `sources/`. The key is a proposal, not proof that its literature distinctions are novel; §13 records the cross-query correction found during this build.

[S13] **FRONT-REUSE-01 protocol, fixtures and results.** Assistant-selected synthetic realization, 5 October 2026. `experiment/FREEZE.json` identifies the protocol and 27 before/after pairs before primary execution; `validity_ledger.json` records expectations. Primary deterministic evidence is `experiment/results_run_01/`. The real-model lane remains `PREPARED_NOT_RUN`. The frozen protocol is within-build predeclaration, not external preregistration.

## C.2 External research

[R1] **Cook, Stephen.** *The P Versus NP Problem*. Official Clay Mathematics Institute problem description. [Primary PDF](#). Checked definitions and NP-completeness discussion. Source for the standard computational question, not a claim that the problem cannot be investigated.

[R2] **Cook, Stephen A., and Robert A. Reckhow.** “The Relative Efficiency of Propositional Proof Systems.” *Journal of Symbolic Logic* 44(1), 36–50, 1979. DOI [10.2307/2273702](#). [Author-hosted PDF](#). Checked §1 definitions and propositions and §3’s sharing of derivations under the same assumptions.

[R3] **Krajíček, Jan.** “The Cook–Reckhow definition.” 2019. [arXiv:1909.03691v1](#). Checked the proof-system definition and polynomial-boundedness/complementation relationship. Used as an author research exposition, not independent verification of FRONT.

[R4] **Marques-Silva, João P., and Karem A. Sakallah.** “GRASP—A New Search Algorithm for Satisfiability.” *Proceedings of ICCAD*, 220–227, 1996. [Primary conference-paper copy](#). Checked conflict analysis, learned causes, and pruning. No reproduction of the paper’s benchmark tables is claimed.

[R5] **Doyle, Jon.** “Truth Maintenance Systems for Problem Solving.” *Proceedings of IJCAI*, p. 247, 1977. [Primary proceedings page](#). Checked this one-page summary of MIT AI Lab report TR-419. It is not cited as if the inaccessible 1979 journal article had been fully inspected.

[R6] **de Kleer, Johan.** “An Assumption-based TMS.” *Artificial Intelligence* 28, 127–162, 1986. DOI [10.1016/0004-3702\(86\)90080-9](#). [Author-hosted scan](#). Checked context dependencies, nogoods, and §6 implementation costs. The scan’s inverted first page was inspected as an image; relevant later passages are readable.

[R7] **Darwiche, Adnan, and Pierre Marquis.** “A Knowledge Compilation Map.” *Journal of Artificial Intelligence Research* 17, 229–264, 2002. DOI [10.1613/jair.989](#). [Paper PDF](#). Checked introduction and definitions of representation/query trade-offs. The 2011 arXiv deposit date is not substituted for the 2002 publication year.

[R8] **Russell, Stuart.** “Rationality and Intelligence.” *Proceedings of IJCAI*, 950–957, 1995. [Primary PDF](#). Checked §§5–6 on metareasoning costs, regress, and bounded optimality. Specific efficiency improvements reported there are not attributed to FRONT.

[R9] **Atserias, Albert, Johannes Klaus Fichte, and Marc Thurley.** “Clause-Learning Algorithms with Many Restarts and Bounded-Width Resolution.” *Journal of Artificial Intelligence Research* 40, 353–373, 2011. [Paper](#)

[PDF](#). Checked abstract and introduction for the restricted algorithm and parameter-dependent proof-complexity result. The later repository deposit is not the publication date.

[R10] Cornell University, CS786, *Introduction to Kleene Algebra*, Lecture 9: “Completeness.” 23 February 2004. [Course-source PDF](#). Checked Lemma 9.2 and the right-congruence/state-quotient conditions. Used as university-authored mathematical exposition. The original 1958 Nerode paper was not fully inspected in this review.

[R11] Clay Mathematics Institute. “P vs NP.” [Official problem-status page](#). Revisited on 4 October 2026; the page labels the problem unsolved. Used for the status sentence in §12, not as a hardness proof. Historical access reports from earlier drafts remain in their source notes.

[R12] Boyd, Stephen. *Observability and State Estimation*, EE263 Lecture 19. Stanford University, Autumn 2008–09. [University-hosted lecture](#). Inspected the known-model/noiseless setup, stacked output map and ambiguity kernel, Cayley–Hamilton stabilization, and invariance, especially lecture pp. 19–4–19–7 and 19–12. Source for established observability placement, not independent verification of FRONT. The proofs here use the manuscript’s notation.

[R13] Levoy, Marc, and Pat Hanrahan. “Light Field Rendering.” Proceedings of SIGGRAPH 1996, corrected July 1996. [Author-hosted paper](#). Read §§1–2 and inspected the two-plane representation in Figure 1. Used only as the neighboring indexed-view representation in §7.2, with static-scene, fixed-illumination, and unobstructed-space assumptions retained. No optical experiment is reported.

### Newly inspected comparisons for v0.4

[R14] Mokhov, Andrey, Neil Mitchell and Simon Peyton Jones. “Build Systems à la Carte.” ICFP 2018, *Proceedings of the ACM on Programming Languages* 2, article 79. [Author-hosted paper](#); expanded JFP 2020 version: DOI [10.1017/S0956796820000088](#). Inspected ICFP §§4.2.2–4.3, printed pp. 79:14–79:16, including the trace formulas. Conference/journal research; no benchmark replication here.

[R15] Acar, Umut A. *Self-Adjusting Computation*. Carnegie Mellon University PhD thesis, 2005. [University-hosted thesis](#). Inspected abstract, overview and trace-stability discussion. Thesis; not a new FRONT method.

[R16] Nadel, Alexander, Vadim Ryvchin and Ofer Strichman. “Ultimately Incremental SAT.” SAT 2014. [Author-hosted paper](#). Inspected §1 and the retained-derivation discussion before §4. The key’s MiniSat ESCAR PDF could not be retrieved; this primary paper supplies the direct incremental-assumptions comparison instead.

[R17] Li, Pengyuan, Boris Glavic, Dieter Gawlick, Vasudha Krishnaswamy, Zhen Hua Liu, Danica Porobic and Xing Niu. “In-memory Incremental Maintenance of Provenance Sketches.” EDBT 2026, pp. 56–68. [Proceedings paper](#); [extended preprint](#), first submitted May 2025. Inspected §§1–2 and Figure 2. The earlier preprint predates the 2026 conference: it must not be described as an idea first appearing in 2026.

[R18] Wu, Michael, and Arquimedes Canedo. “Invalidation Contracts for Cross-Episode Agent Memory.” [arXiv:2609.00243v1](#), submitted 31 August 2026. Inspected abstract, §§3–4 and limitation headings. Recent preprint; reported model results are not independently reproduced here.

[R19] Wang, Yiqi, et al. “PACMI: Provenance-Aware Cascading Memory Invalidation for Long-Term LLM Agents.” [SSRN 7480138](#), DOI [10.2139/ssrn.7480138](#). Written 17 September, posted 19 September 2026. Recent preprint. Only the indexed primary abstract/metadata was retrieved; full-text requests failed. No detailed numerical or theorem comparison is claimed.

[R20] Wu, Ming, and Pengyuan Zhu. “Agent Zero Memory: Provenance-Aware Long-Term Memory for LLM Agents.” [arXiv:2608.29606v1](#), submitted 30 August 2026. Inspected abstract, §3.1 and limitations. Recent preprint; no independent result replication. Its citation restrictions and semantic-support claims remain distinguishable in this comparison.

[R21] Acar, Umut A., Matthias Blume and Jacob Donham. “A Consistent Semantics of Self-Adjusting Computation.” ESOP 2007, pp. 458–474, DOI [10.1007/978-3-540-71316-6\\_31](#); [author proof page](#), later

[arXiv:1106.0478](https://arxiv.org/abs/1106.0478). Publisher metadata and primary abstract inspected. Twelv resources are linked by the authors; their proofs were not rerun here.

All new access reports refer to 5 October 2026. The source map separates successful full-text access, section-level inspection and abstract-only comparison.

### C.3 Authorship and provenance

Christopher D. Pang is the author of FRONT. He originated its name and expansion and supplied its conceptual direction and corpus. OpenAI ChatGPT, used as Hearthline, assisted with retrieval, literature review, mathematical formulation, proof drafting, executable checks, document preparation, and editing. Its outputs are proposals and tool results, not independent corroboration or author adoption.

The supporting materials retain the source identities, protocols, fixtures, results and formal receipts needed to inspect the claims. Earlier evidence remains attributable to its original version. The original v0.1 evidence gap remains labeled. A hash verifies bytes, not mathematical authority.

#### v1.0 source additions and retrieval scope

[S14] **Origin and activated key, 8 October 2026.** Chris Pang’s supplied conversation and FRONT\_v1.0\_Research\_and\_Build\_Key.md, with the earlier unadopted v0.5 thinking key. Source hashes and bounded contribution descriptions are recorded in SOURCE\_MAP\_v1.0.json. The original private conversation and full handoff keys are not redistributed. The reported phrase “Correction needs its own mechanism” is attributed to the separate episode as reported by Pang, not as a recovered independent proof.

[S15] **PAL Single-Cut Transport Lemma, local v0.1, 11 August 2026.** Christopher D. Pang, supplied source-derived manuscript, §§3–5. Exact file identity is retained. User-supplied public reference: <https://zenodo.org/records/22239108>. The public/local version and byte match remain unverified following retrieval attempts. The note attributes its underlying slot/interface/OR construction to the deposited source at <https://zenodo.org/records/21871667>; FRONT imports only the displayed finite local count and OR example, not the global NP-completeness or priority claims.

[S16] **FRONT v1.0 retained build and FRONT-CORRECT-01, 8 October 2026.** Assistant-selected finite realization under Pang’s activated key. FREEZE.json identifies the reconstructed implementation/protocol/75 fixtures before the retained comparison. results\_retained\_01 contains the primary events; results\_retained\_optimized is a robustness rerun, not extra independent observations. Original v0.4 evidence remains in experiment\_reuse\_v0.4. Review independence is limited to separate algorithms and exact small-fixture enumeration, not separate authors or a second Lean kernel.

[S17] Pang, Christopher D. FRONT v1.0 Lean audit companion. 8 October 2026. <https://github.com/Grativy6/front-lean>. Audit/COVERAGE.md maps all 21 propositions; Audit/RECEIPT\_BOOK.md reports assumptions, finite experiments and remaining bridges. The original formal receipt checks 163 declarations and 13 modules under Lean 4.32.1 and Mathlib commit 520045ab14e26149ee970e2e617ca04b09bde5d6. Publication/source-concordance.json binds the editorial release to the unchanged mathematical source and proof inputs.

[R22] **Alchourrón, Carlos E.; Gärdenfors, Peter; Makinson, David.** “On the Logic of Theory Change: Partial Meet Contraction and Revision Functions.” *Journal of Symbolic Logic* 50(2), 510–530 (1985). DOI: 10.2307/2274239. Direct comparison concerns contraction/revision and selection, not a claim to replace the AGM postulates.

[R23] **Arenas, Marcelo; Bertossi, Leopoldo; Chomicki, Jan.** “Consistent Query Answers in Inconsistent Databases.” *PODS* (1999), 68–79. DOI: 10.1145/303976.303983. Author-hosted copy: <https://www.cs.toronto.edu/~marenas/publications/pods99.pdf>. The minimal-repair / consistent-answer account is compared with finite recorded-support existence; their quantifiers differ.

[R24] **Green, Todd J.; Karvounarakis, Grigoris; Tannen, Val.** “Provenance Semirings.” *PODS* (2007), 31–40. DOI: 10.1145/1265530.1265535. The comparison uses the authors’ presentation at

<https://www.cis.upenn.edu/~plclub/propr/greg-slides.pdf> for joint and alternative provenance. It is not represented as a new complete audit of the paper.

R5/R6's truth-maintenance ancestry is retained. Primary-source reading descriptions from the earlier attempt inform this reconstruction; an interrupted run is not a new independent reviewer. No copyrighted reference PDF is redistributed. The original v0.4 citation checks remain historical reports, not claims that every source was re-opened during this continuation. Novelty or priority is not concluded from this bounded comparison.

## Appendix D. Proofs and boundaries of the linear continuation adapter

This appendix completes the arguments stated in §9. It uses exact vector-space maps, not a noise model. The algebra is elementary or inherited as attributed in the main text. Its role is to make the BRIDGE-FRONT application inspectable within this manuscript.

### D.1 Proof of Proposition 5B

Let  $K = \ker r$ . If  $L(K) \subseteq K$ , induction gives  $L^j d \in K$  for all  $d \in K$  and  $j \geq 0$ , hence every readout is zero. Conversely, the condition at  $j = 1$  gives  $L(K) \subseteq K$ . This quantifies over the full linear difference space; for a restricted state set it can be stronger than necessary.

The kernel of  $O_N$  is the intersection of its component kernels, so  $K_{N+1} \subseteq K_N$ . The characteristic polynomial of  $L$  has degree  $n$ ; Cayley–Hamilton expresses  $L^n$  as a linear combination of  $I, L, \dots, L^{n-1}$ . By further multiplication, every higher power is in that span. Any vector annihilated by  $r L^j$  for  $0 \leq j < n$  is therefore annihilated for all  $j$ , proving  $K_n = K_\infty$ . If  $d \in K_\infty$ , then  $r L^j(L d) = r L^{j+1} d = 0$ , so the limiting space is invariant. Finally, equal future outputs of  $x, x'$  mean exactly that every  $r L^j(x - x')$  vanishes. ■

For several update letters, let  $L_w$  apply the letters of  $w$  in execution order; thus  $L_{uv} = L_v L_u$ . If every finite word and every specified view after it is included, then  $r_f L_w L_a d = r_f L_{aw} d = 0$  for each invisible  $d$ . This proves invariance under each  $L_a$ . Restricting the admitted pairs can remove  $(f, aw)$  and invalidate this argument. No single finite observed path supplies all-word coverage by itself.

A simple restricted-domain control uses  $L(x, y) = (x + y, y)$ ,  $r(x, y) = x$ , and states only  $(0, 0), (1, 0)$ . The ambient kernel is not invariant, but the present readout already distinguishes both admitted states. The necessary condition is about admitted pairs, not every ambient vector.

### D.2 Proof of Proposition 5D and refinement compatibility

Set  $K = \ker J$  and  $B = \Im e$ . Define

$$\begin{aligned} i: K / (B \cap K) &\rightarrow T / B, i(k + (B \cap K)) = k + B, \\ \acute{J}: T / B &\rightarrow J(T) / J(B), \acute{J}(x + B) = Jx + J(B). \end{aligned}$$

Both maps are well-defined. The first is injective because  $k \in K$  maps to zero exactly when  $k \in B \cap K$ . The second is surjective by its codomain. Its kernel consists of classes  $x + B$  with  $Jx = Jb$  for some  $b \in B$ , equivalently  $x - b \in K$ . This kernel is exactly the image of  $i$ . Exactness and the identification  $T / (B + K) \cong J(T) / J(B)$  follow. The two short sequences for  $B \cap K \subseteq B$  and for (20j) give the four-sector dimension identity. ■

If a refined readout  $J'$  retains all components of  $J$ , a coordinate projection  $p$  satisfies  $J = p J'$ . Therefore  $K' \subseteq K$ . The induced map  $K'/(B \cap K') \rightarrow K/(B \cap K)$  is injective; the map on  $T/B$  is the identity; and  $[J'x] \mapsto [Jx]$  is a well-defined surjection between the visible quotient spaces. These maps commute with the exact sequences. Refinement makes a smaller hidden sector and larger visible sector within the same generative defect; it does not create new generated elements.

More generally, BRIDGE's supplied frame-morphism conditions are  $Pe = e'u$  and  $J'P = CJ$ . They give  $P(B) \subseteq B'$  and  $P(K) \subseteq K'$ . The same  $P$  then induces maps on  $B \cap K$ ,  $B/(B \cap K)$ ,  $K/(B \cap K)$ , and  $T/(B+K)$ , by taking each representative to its image. The commuting squares certify the transport; an arbitrary relabeling without those relations does not. [S9, "Frame Morphisms and Certificate Transport"]

The common-baseline example in §9.7 is an important limitation on implementation. The quotient  $J(T)/J(B)$  removes one vector  $Jb$  for a single  $b \in B$ . The product of separate per-view quotients instead allows one  $b_f$  per view. These need not be the same equivalence relation.

### D.3 Proof of Proposition 5E

Write  $K = \ker J$ . The joint readout  $(J, s)$  has kernel  $K \cap \ker s$ . Linear factorization gives (20k).

For any repairing  $s$ , the restriction  $q|_K$  factors through  $s|_K$ . Hence

$$\text{rank}(q|_K) \leq \text{rank}(s|_K) \leq \text{rank } s \leq \dim E. \quad (\text{D1})$$

This proves the rank and codomain lower bounds. If  $\dim E \geq d_q$ , choose a linear injection  $a: q(K) \rightarrow E$ . Set  $s(k) = a(qk)$  on  $K$ , and extend  $s$  by zero on a chosen complement of  $K$  in  $T$ . Then  $\text{rank } s = d_q$ , and  $s(k) = 0$  implies  $qk = 0$  for  $k \in K$ . This attains the bound and satisfies (20k). The zero-query case permits the zero side trace. **I**

For the residual-target lane apply the same proof with state space  $D = T/B$ , output  $J'$ , and query  $q_D$ . Its kernel is  $i(H)$  by Proposition 5D, so the required rank is  $\text{rank}(q_D \circ i)$ . A side trace on  $D$ , pulled back to  $T$ , annihilates  $B$ . A state query not annihilating  $B$  cannot be treated as a query on  $D$ .

For finite matrix representations there is a useful independent identity:

$$\text{rank}(q|_{\ker J}) = \text{rank} \begin{pmatrix} J \\ q \end{pmatrix} - \text{rank } J. \quad (\text{D2})$$

Indeed, the left side is  $\dim \ker J - \dim(\ker J \cap \ker q)$ ; rank-nullity gives the right side. The companion uses this identity, an explicit side-trace extension, a solved decoder, and a separate two-column minor calculation in its bounded cases. Agreement among these checks is not a proof for all inputs; the arguments above are the all-domain manuscript basis.

### D.4 Proof of Proposition 5F and coordinate equivalence

Suppose  $L(B) \subseteq B$ . Representatives differing by  $b \in B$  have images differing by  $Lb \in B$ ; therefore  $[x] \mapsto [Lx]$  is well-defined and linear. Conversely, if the formula is well-defined,  $[b] = [0]$  forces  $[Lb] = [0]$  for every  $b \in B$ , so  $L(B) \subseteq B$ . Replacing  $B$  by  $K$  gives the readout-quotient condition. **I**

A compatible invertible coordinate change does not add information. With  $L' = PLP^{-1}$  and  $r' = CrP^{-1}$ , induction gives  $(L')^j P = PL^j$ , and hence

$$r'(L')^j(Px) = CrL^jx. \quad (D3)$$

The output changes by an invertible re-expression, so equalities and distinctions are preserved. Reordering stored labeled observations similarly preserves the joint data. Reordering actual noncommuting updates, however, need not preserve the state. These are distinct operations with different obligations.

## D.5 Delayed-readout examples retained from v0.3 §9.3

First keep the view fixed. On  $T = Q^3$ , let

$$U(x, y, z) = (x + y, y + z, z), r(x, y, z) = x, \alpha = (0, 0, 1). \quad (20d)$$

The three states are  $(0, 0, 1)$ ,  $(0, 1, 1)$ , and  $(1, 2, 1)$ . Their first-coordinate readings are  $0, 0, 1$ . The first discrepancy from the initial state is  $(0, 1, 0)$ : nonzero but hidden by  $r$ . The next discrepancy,  $(1, 2, 0)$ , is readable. The update, rather than an unspecified influence, supplies the dependency through which the hidden direction reaches the readout.

For an arbitrary initial state, the three retained readings satisfy

$$\begin{pmatrix} s_0 \\ s_1 \\ s_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (20e)$$

Consequently  $x = s_0$ ,  $y = s_1 - s_0$ , and  $z = s_2 - 2s_1 + s_0$ . The successive joint kernels have dimensions  $2, 1, 0$ . This is exact recovery from the *joint* readings under a known update. The initial value  $s_0$  alone does not produce the other two readings.

Now choose  $U_0 = \text{diag}(1, 2, 3)$  with the same  $r$ . The difference  $(0, 1, 0)$  changes under iteration, but every reading of it is zero. There is genuine evolution with persistent invisibility to this interface. An extra second-coordinate sensor reveals it immediately. These are different routes: evolving the state under a fixed readout, and changing what the readout distinguishes.

## D.6 Proof of Proposition 5C

**Proof.** Choose bases in the codomains and consider the scalar component rows as elements of  $T^*$ . Their span has a basis selected from these rows with at most  $n$  elements. Select the readouts containing those rows. A vector is annihilated by the selected readouts exactly when it is annihilated by the full row span, which is exactly the full common kernel. ■

## Appendix E. Finite correction record and reproducibility

### E.1 What a packet carries

A passive correction packet identifies the claim event, selected derivation and root witnesses, typed discrepancy, interpretation/version/scope, recorded affected uses, alternative proof handles, unresolved work, permitted next operation or STOP, and receipt location. A receiving desk may inspect it without inheriting the sender's authority. This is a proposed Branchline/Pony Post interface; the finite cross-desk fixture is not a deployment test.

The machine-readable fixtures use sources, roots, conjunctive rules, cached selected-proof IDs, history, grants, requests and queries. They instantiate the existing state account rather than establish a universal data schema. Answers include supported, proof and route; unresolved answers retain no fabricated proof. Outputs separately include

diagnosis, intent NOT\_ESTABLISHED, coverage RECORDED\_EDGES\_ONLY, history, action-permission tokens, stop, operations and timings.

## E.2 Mandatory family coverage

| Family | Retained test                                                                          |
|--------|----------------------------------------------------------------------------------------|
| 1      | Refused SCT attempt; separate 96-case row-budget assay                                 |
| 2      | Contrary assertion stays in operational history                                        |
| 3      | Only unsupported selected route                                                        |
| 4      | Conjunctive independent alternative survives                                           |
| 5      | Same answer, changed derivation                                                        |
| 6      | Changed source scope/version, fresh witness ID                                         |
| 7      | No recorded alternative, honest unresolved                                             |
| 8      | Timeout, missing evidence, negative, missing grant, tool error remain typed            |
| 9      | Alternate method to the same denied effect                                             |
| 10     | One hundred assertions do not supply warrant                                           |
| 11     | Correct hash/untrusted source, expired grant, wrong interpretation, changed dependency |
| 12     | Tom/Speck packet, historical attribution, no grant inheritance                         |
| 13     | Internal inspection allowed, public disclosure denied                                  |
| 14     | Grounded/ungrounded cycles and omitted root                                            |
| 15     | Candidate repair violates another condition and rolls back in scratch                  |
| 16     | Quiet completed task, STOP without a new task                                          |

All complete case records, the family/scale index and input hashes are shipped. The fixture list is finite and designed; it is not a representative sample of deployed agents.

## E.3 Rerun commands

From the package root, using Python 3.10 or newer and no third-party libraries:

```
python scripts/verify_package.py
python experiment_correct/run_correct.py --output results_my_replay
python -O experiment_correct/run_correct.py --output results_my_optimized_replay
python scripts/recheck_legacy.py
```

Use a new result-directory name for each correction run. Existing results and frozen fixtures cannot be overwritten by the runner. Reruns add files and may generate bytecode caches; the package manifest verifies its named immutable members, not an assertion that a directory can never gain files. Python -O does not remove the explicit runtime checks. Timings need not match.

## E.4 Trust and remaining work

The frozen program, Python interpreter, operating system and hardware remain part of the execution trust base. The separate fixed-point oracle is additionally compared with explicit valuations in 1,024 two-atom systems, but no general implementation-correctness theorem follows. Truth of source metadata, complete capture of external effects, dynamic concurrency, live authentication, durable rollback, deception attribution and practical agent costs are not verified. The independent-review packet and claim ledger identify those boundaries. The real-model lane remains unrun. The separately checked Lean companion is described in §17.2; no general Python-to-Lean refinement is claimed. [S17]